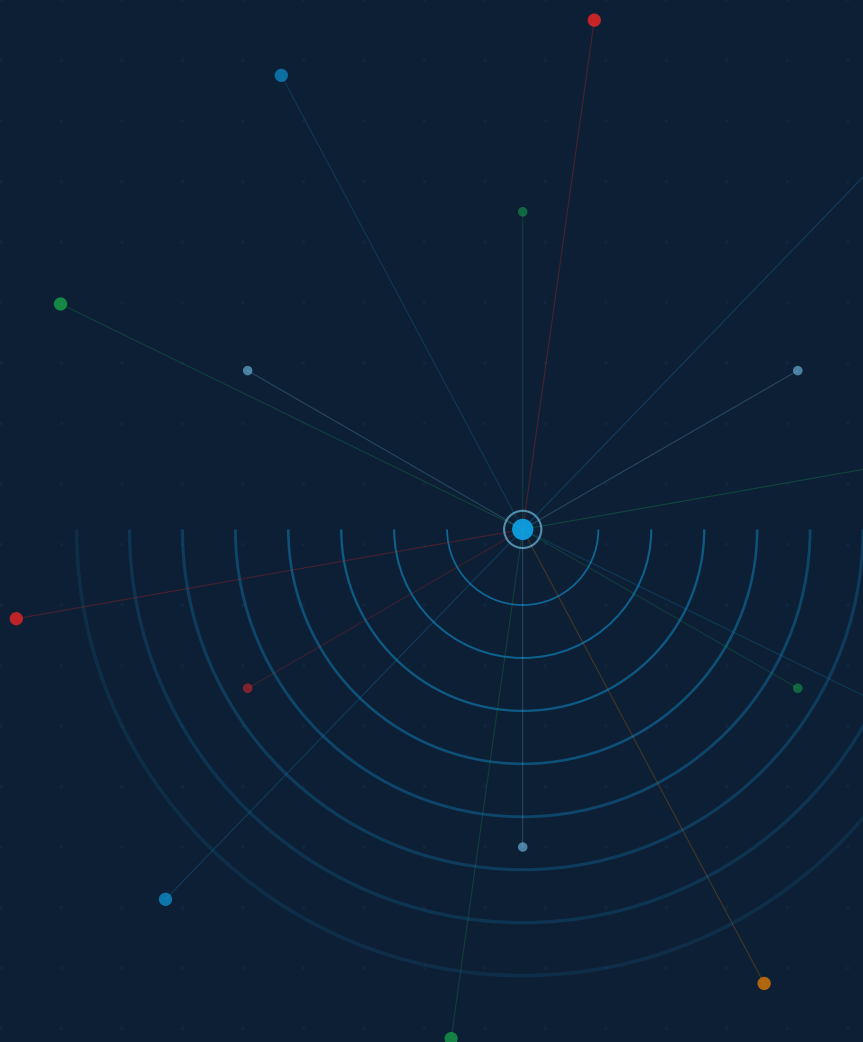


JA4proxy

Enterprise TLS Security Proxy

Technical Reference Manual

Complete reference for architects, developers, and security engineers



JA4proxy Documentation Series

<i>Product Overview</i>	Introduction for buyers and evaluators
<i>Operator's User Guide</i>	Installation, configuration, and day-to-day operations
<i>Technical Reference Manual</i>	This document — complete specification for all components

Status: POC — Functional and tested. Not yet production-hardened.

Source: <https://github.com/seanpor/JA4proxy> JA4 spec: <https://github.com/FoxIO-LLC/ja4>

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System Architecture

JA4proxy makes allow/block/tarpit decisions based entirely on plaintext meta-data visible before and during the TLS handshake. It never decrypts traffic, never holds TLS keys, and forwards allowed connections byte-for-byte unchanged.

THREE DESIGN PRINCIPLES govern every architectural decision in JA4proxy: **fail open** (a missed threat is recoverable; a blocked legitimate user is not), **never decrypt** (the proxy is cryptographically transparent — it reads only the plaintext ClientHello), and **monitor first** (the default dial is 0; the system observes and scores before it ever blocks).

Design Principles

The Asymmetry Principle

The most important design constraint in JA4proxy is the cost asymmetry between the two error types:

Error type	Example	Cost	Consequence
False negative	Bad bot slips through during cache sync window	Low	Recoverable. Log entry exists; retrospective analysis possible; no user harm.
False positive	Real browser is blocked or tarpitted	High	Not recoverable. User leaves. Revenue impact. Reputation damage.

Table 1: Error cost asymmetry — the governing principle of all threshold and TTL decisions.

Every caching TTL, every threshold default, every fallback behaviour, and every new feature decision flows from this asymmetry. When the answer is not obvious, fail open.

The Asymmetry Rule

ALLOW decisions are cached for 30 minutes. BLOCK decisions are cached for 30 seconds. When Redis says block but the local in-process cache says allow, the local cache wins. If Redis is down, real browsers continue to work.

Never Decrypt

JA4proxy operates entirely on **plaintext metadata**. The TLS ClientHello — the first message a TLS client sends — is transmitted in plaintext before any encryption is established. It contains:

- The TLS version and list of supported cipher suites
- Supported extensions and their parameters
- The SNI (Server Name Indication) hostname
- ALPN (Application-Layer Protocol Negotiation) values
- Elliptic curve groups and point formats

From these fields, the JA4 fingerprinting algorithm produces a deterministic string that identifies the TLS client implementation (browser, bot, scanner, malware C2) with high accuracy. The proxy never participates in the TLS handshake — it is a passthrough observer and gatekeeper, not a terminator.

Monitor First

The default configuration ships with `dial: 0`. At `dial=0`, the pipeline runs completely — all signals are collected, all scores are computed, all decisions are made — but no connection is ever blocked. Every connection is allowed with a `MONITOR` annotation in the log.

This means operators can deploy JA4proxy in front of production traffic immediately, observe the score distribution, tune thresholds, and build confidence before enabling any blocking. Raising the dial is a conscious operator decision.

Config-Driven, Hot-Reloadable

Every behavioural parameter is configurable in `config/proxy.yml`. The proxy watches for `SIGHUP` and reloads configuration without dropping connections. The Management UI triggers reload via a Redis pub/sub message. After reload, new configuration applies to all subsequent connections.

Only the listen port, Redis URL, and TLS certificate paths require a process restart. All other parameters — dial, bypass conditions, thresholds, blacklists, whitelists — are hot-reloadable.¹

¹ Config reload events are logged and reflected in the `ja4proxy_config_reloads_total` Prometheus counter.

Component Overview

Network Topology

Traffic Flow Diagram

Network Zones

A production deployment uses four network zones:

Component	Language	Port	Responsibility	Table 2: System components and their responsibilities
HAProxy	C	:443	TLS passthrough load balancer; PROXY protocol v2 injection; upstream TLS listener	
JA4proxy Proxy	Go*	:8080	TLS ClientHello sniffing; pipeline execution; allow/block/tarbit/ban decisions; Redis I/O	
Redis	Redis 7	:6379	Shared state: bans, rate windows, beaconing timestamps, JA4 lists, HLL per CIDR, event stream	
Analytics Node	Python	—	Reads Redis Streams (XREADGROUP); cross-instance aggregation; campaign detection; writes findings to Redis	
Prometheus	Go	:9090	Metrics collection and storage	
Grafana	Go	:3000	Dashboards; visualisation of proxy telemetry	
Loki	Go	:3100	Structured log aggregation	
Alertmanager	Go	:9093	Alert routing and deduplication	
Backend	any	:443	Protected origin service; receives only allowed, unmodified TLS connections	

*The Go proxy (`cmd/proxy/`, `internal/`) was promoted to production in Phase 15 and is the only implementation that ships in releases, Docker images, and Helm charts. The Python proxy (`proxy.py`, `src/security/`) is retained as an experimental prototyping surface for new signal modules; production-runtime Python is limited to the Management API, analytics node, and compliance reporting — services that are not on the proxy hot path.

The Backend origin server sits behind the App zone, reachable only from JA4proxy instances. Internet traffic never reaches the App zone directly. The Data zone is not routable from the DMZ or from the internet.

Data Flow: TLS Connection Lifecycle

Stage 1 — TCP Accept and IP Extraction

When HAProxy accepts a TLS connection, it wraps it in a PROXY protocol v2 header² and forwards the raw TCP stream to a JA4proxy instance on port 8080.

JA4proxy reads the PROXY protocol header first, extracting:

- The **real client IP** (IPv4 or IPv6)
- The real client port
- The HAProxy (sender) IP — used for upstream CIDR trust verification

² PROXY protocol v2 is a binary framing format prepended by HAProxy to the TCP stream. It carries the original client IP:port and the HAProxy IP:port.

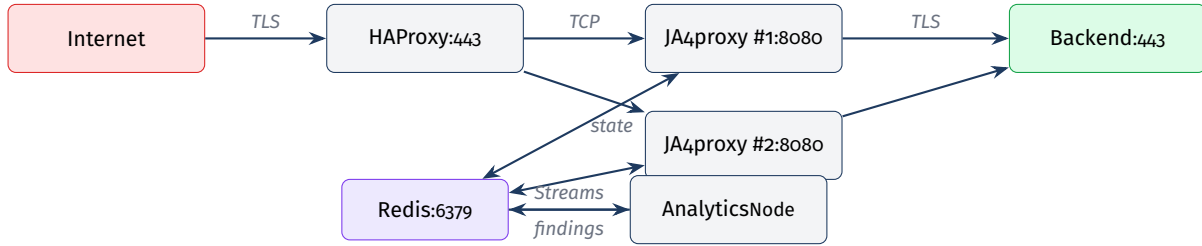


Figure 1:
HAProxy
headers so
client IP.

Zone	Subnet (example)	Contains	Permitted inbound traffic
DMZ	10.0.1.0/24	HAProxy load balancer	Internet :443 only
App	10.0.2.0/24	JA4proxy proxy instances	HAProxy TCP from DMZ only
Data	10.0.3.0/24	Redis, analytics node	App zone on :6379 only
Management	10.0.4.0/24	Prometheus, Grafana, Loki	Operator VPN only

The client IP is canonicalised to its compressed string representation (`ipaddress.ip_address(ip).compressed` in Python, `netip.Addr.String()` in Go) immediately. All subsequent pipeline steps use this canonical form.

Stage 2 — TLS ClientHello Sniff

After PROXY protocol extraction, JA4proxy reads the first TLS record from the TCP stream. This is always the ClientHello — a plaintext record (record type 0x16, content type 0x01). The proxy parses:

- Handshake type and TLS record version
- Client random (32 bytes, not used for fingerprinting)
- Session ID length and value
- Cipher suite list (ordered)
- Compression methods
- Extensions: SNI (0x0000), ALPN (0x0010), supported groups (0x000a), EC point formats (0x000b), signature algorithms (0x000d), supported versions (0x002b), PSK key exchange modes (0x002d), key share (0x0033), and others

From this data, the JA4, JA4X, and JA4T fingerprints are computed (see Chapter). The entire sniff operation is non-blocking and completes before any response is sent.

Stage 3 — Pipeline Execution

The pipeline (Chapter) runs all 12 layers. Layers 0–2 are bypass/block decisions that short-circuit the pipeline immediately. Layers 4–8 are signals that run in parallel via `asyncio.gather()`. The pipeline produces a score (0–100) and an action.

Stage 4 — Action Execution

Based on the action from the pipeline:

ALLOW The buffered ClientHello bytes are forwarded to the backend. JA4proxy acts as a transparent TCP relay for the remainder of the connection.

BLOCK A TCP RST is sent. The connection is closed immediately.

TARPIT The connection is accepted but data is delivered to the backend at an artificially low rate (or deliberately delayed), consuming attacker resources.

BAN The IP is added to Redis with a TTL. Subsequent connections from this IP are rejected immediately at layer 0 without pipeline execution.

Flag The connection is allowed but a high-priority alert is emitted to the Redis Stream for the Analytics node.

Stage 5 — Event Emission

Every connection that completes pipeline evaluation produces a JSON event written to the `events` Redis Stream via `XADD` (fire-and-forget). The analytics node consumes these events for cross-instance aggregation, campaign detection, and score drift alerting. Stream writes add zero latency to the hot path.

IPv6 Handling

IPv6 is a first-class concern throughout JA4proxy. Every feature that touches an IP address handles both address families:

Subsystem	IPv4 rule	IPv6 rule
Rate limiting keys	Full IP string	Full IP string (no truncation)
CIDR matching	In-process pytricia / Go trie	Same trie handles both natively
GeoIP lookup	MaxMind GeoLite2 City/ASN	Same DB covers IPv6
HyperLogLog per subnet	<code>hll:cidr24:{cidr} (/24)</code>	<code>hll:cidr48:{cidr} (/48)</code>
Beaconing key	<code>beacon:{ip}:{ja4}</code>	Same key format, full IPv6
Ban key	<code>ban:{ip}</code>	Same key format, full IPv6
RDAP lookup	RIR via IANA bootstrap	Same bootstrap — works for IPv6
Block expansion	Never beyond /24	Never beyond /48

Table 4: IPv6 handling rules by subsystem

IPv6 /48 Equivalence

A /48 IPv6 prefix is the approximate equivalent of an IPv4 /24 for population density purposes. A typical ISP assigns a /48 to a single customer premises. Block expansion (RDAP Phase 11) never expands beyond /48 for IPv6 or /24 for IPv4 — expanding further would risk blocking entire ISP customer populations.

The Go and Python Implementations

Go (Production Runtime)

The Go proxy in `cmd/proxy/` and `internal/` (built to `bin/proxy`) is the production runtime. It was promoted to production in Phase 15 and is the only implementation that ships in releases, Docker images, Helm charts, and enterprise documentation. The Go implementation uses `net/netip` for IP handling, a `crypto/tls`-derived `ClientHello` parser, and `goroutine-per-connection` concurrency. It targets 10–50× the throughput of the Python prototype.

Python (Experimental Prototyping Surface)

The Python implementation in `proxy.py` and `src/security/` is retained as an experimental prototyping surface for new signal modules. Engineers prototype in Python, prove the idea, then port it to Go before it ever touches real traffic. Python services that are *not* the proxy — the Management API (FastAPI), the analytics node, and the compliance reporting service — remain production Python because they are not on the connection-handling hot path.

Implementation Status

The Go proxy is the only path that handles production traffic. The Phase 200-series hardening (Redis TLS, PROXY v2 with TLVs, default-credential removal, expanded signal coverage) lands in the Go runtime first; Python is updated only when needed for prototyping.

Horizontal Scaling

Multiple JA4proxy instances share state through Redis. Because all ban decisions, rate windows, JA4 lists, and beaconing data live in Redis, any instance can enforce decisions made by any other instance. There is no sticky routing requirement — HAProxy distributes connections across instances with standard round-robin or least-connection load balancing.

The only per-instance state is the in-process LRU cache (`src/-cache/local_cache.py`). This cache is intentionally not synchronised across instances. Cache misses are handled by Redis fallback. The local cache uses conservative TTLs (30min for ALLOW, 30s for BLOCK) to bound the window in which a newly added ban is not yet enforced by a particular instance.

The Security Pipeline

The pipeline is the ordered sequence of checks applied to every TLS connection. Every layer runs in microseconds. Signal collection layers run in parallel. The pipeline produces exactly one action per connection.

THE PIPELINE HAS THREE ZONES: the **fast-path bypass zone** (layers 0–2), which short-circuits immediately on certain conditions; the **signal collection zone** (layers 3–10), where eight checks run concurrently; and the **decision zone** (layers 11–12), which aggregates signals into a score and maps the score to an action.

Pipeline Overview

Table 5: Complete pipeline layer reference				
Layer	Name	Zone	Output	Trigger condition
o	IP trust & normalisation	Fast path	—	Every connection
ob	Static IP allowlist	Fast path	ALLOW	IP in static allowlist
oc	GeoIP country block	Fast path	BLOCK	Country in blacklist
od	CIDR block	Fast path	BLOCK	IP in blocked subnet
oe	Spamhaus DROP/EDROP	Fast path	BLOCK	IP in DROP/EDROP trie
1	ALPN browser bypass	Fast path	ALLOW	ALPN = h2 or h1
1b	JA4 whitelist	Fast path	ALLOW	JA4 in whitelist SET
1c	mTLS client cert	Fast path	ALLOW	Valid client certificate
2	JA4 blacklist	Fast path	BLOCK	JA4 in blacklist SET
3	TLS enforcement	Signal	Signal or BLOCK	TLS version/cipher check
4	SNI analysis	Signal	Signal	Every connection
5	TCP & connection behaviour	Signal	Signal	Every connection
6	ASN classification	Signal	Signal	Every connection
7	FCrDNS enrichment	Signal	Signal	Every connection
8	Beaconing detector	Signal	Signal	Every connection
9	AbuseIPDB score	Signal	Signal	Every connection
10	RDAP org reputation	Signal	Signal	Every connection
11	Attacker attribution	Analysis	Signal	Cross-IP JA4/JA4X correlation
12	Behavioural analysis	Analysis	Signal	Sequential/coordinated patterns
13	Risk scorer	Decision	Score 0–100	Always
14	Action decider	Decision	Action	Always

The Fast-Path Bypass System

How Bypasses Work

Each bypass condition is independently configurable under the `security_policy` configuration section. When a bypass condition is met and its bypass is enabled, the pipeline short-circuits immediately — the signal collection and scoring layers never execute, and no Redis signal-collection writes are made.

All bypass conditions are toggleable. Disabling an ALLOW bypass causes those connections to pass through the full scorer. Disabling a BLOCK bypass routes those connections through the scorer, where the dial controls whether they are actually blocked.

Bypass Modification Risk

The proxy emits a startup `WARN` log entry and increments a Prometheus counter for every high-risk bypass that is disabled. Operators should monitor `ja4proxy_bypass_total` to verify bypass layer traffic volumes are as expected.

Layer o — IP Trust and Normalisation

Every connection enters layer o. This layer:

1. Verifies the connection originates from a trusted upstream CIDR (the HAProxy address range)
2. Reads the PROXY protocol v2 header to extract the real client IP
3. Canonicalises the IP to its compressed string representation
4. Checks whether the IP is in the in-process ban cache (fast reject before Redis)

If the sender is not in the trusted upstream CIDR, the connection is rejected immediately. This prevents IP spoofing via injected PROXY protocol headers.³

³ Without trusted upstream CIDR verification, an attacker who can reach port 8080 directly could inject a PROXY header claiming any client IP.

Layer ob — Static IP Allowlist

Configuration key: `security_policy.static_ip_allowlist.enabled`

If the client IP is found in the static IP allowlist (configured in `config/proxy.yml` and hot-reloadable), the connection is immediately allowed without scoring. Typical use: trusted monitoring systems, CI test runners, internal health checkers.

Layer oc — GeoIP Country Block

Configuration key: `security_policy.country_blacklist_bypass.enabled`

Country lookup uses the IP2Location LITE database in conjunction with a Redis-backed country blacklist. Countries in `geoip.country_`

`blacklist` trigger an immediate BLOCK. This check runs before any TLS parsing — the country decision is based on IP alone.

Country Allowlist vs Blacklist

`geoip.country_whitelist_enabled` and `geoip.country_blacklist_enabled` are independent toggles. The allowlist mode (only listed countries are permitted) is distinct from the blacklist mode (listed countries are blocked). Both can be active simultaneously; the whitelist check runs first.

Layer *od* — CIDR Block

Subnet blocks are maintained in Redis and refreshed every 30 seconds into an in-process trie. This layer is the enforcement point for RDAP block expansion (Phase 11) and manually added subnet bans. Matching an entry produces an immediate BLOCK without scoring.

IPv6: The trie handles both IPv4 and IPv6 prefixes natively. Block expansion never adds prefixes shorter than /24 (IPv4) or /48 (IPv6).

Layer *oe* — Spamhaus DROP/EDROP

Configuration key: `security_policy.spamhaus_bypass.enabled`

Spamhaus DROP and EDROP lists are fetched at startup and refreshed every `spamhaus.refresh_interval` seconds (default 3600). They are stored in an in-process trie for $O(\log n)$ lookup.⁴

When disabled, Spamhaus matches produce a `RiskSignal(+80)` instead of a hard block. The signal flows through the scorer, and the dial controls whether the connection is blocked.

⁴ The Spamhaus DROP list contains IP ranges that have been hijacked or directly allocated to cybercriminals. False positive rate is near zero — these ranges are never used for legitimate traffic.

Layer 1 — ALPN Browser Bypass

Configuration key: `security_policy.alpn_browser_bypass.enabled`

ALPN Bypass — Do Not Disable Without Justification

The ALPN browser bypass is structurally the most critical bypass. `h2` and `h1` ALPN values are set exclusively by real browsers communicating over HTTPS. Disabling this bypass causes all browser traffic to be scored. With typical score distributions, this will generate false positives at scale. This bypass should only be disabled to score specific `h2` API clients, and only with monitoring in place.

Connections presenting `h2` or `h1` ALPN values are immediately allowed. These connections are never enqueued for enrichment, and their IP/netblock is never eligible for block expansion.

Layer 1b — JA4 Whitelist

Configuration key: `security_policy.ja4_whitelist_bypass.enabled`

The JA4 whitelist is a Redis SET (`ja4:whitelist`) loaded into an in-process set at startup and refreshed on config reload. JA4 fingerprints matching the whitelist trigger an immediate ALLOW. Wildcard and regex matching are not supported — whitelist entries are exact fingerprint strings.

Layer 1c — mTLS Client Certificate

Configuration key: `security_policy.mtls_bypass.enabled`

A valid mTLS client certificate is cryptographic proof of identity. By default, presenting a valid certificate against the configured trusted CA set (`config/trusted_cas.pem`) triggers an immediate ALLOW without scoring. If disabled, mTLS clients are scored — they can still be blocked if their score exceeds the dial threshold.

Layer 2 — JA4 Blacklist

Configuration key: `security_policy.ja4_blacklist_bypass.enabled`

The JA4 blacklist is a Redis SET (`ja4:blacklist`) mirroring the whitelist architecture. Matching fingerprints trigger an immediate TCP RST with no further processing. Known blacklist entries include tool and C2 implant fingerprints such as `t13d190900_9dc949149365_97f8aa674fd9` (Sliver C2 framework).

When disabled, blacklisted fingerprints produce a high-weight `RiskSignal` routed through the scorer. This is useful for investigating traffic from known-bad fingerprints without hard-blocking.

Signal Collection (Layers 3–10)

Layers 3–10 run concurrently via `asyncio.gather()` after the fast-path zone completes without a decision. Each signal check is a coroutine that:

1. Reads relevant data from the connection context and Redis
2. Optionally calls an external service (AbuseIPDB, RDAP, DNS)
3. Returns a `RiskSignal(score, weight, reason)` object

All signal checks are non-blocking. External service calls are fire-and-forget via `asyncio.create_task()` — they do not block the connection handling coroutine. Every signal check has an explicit failure handler that logs the failure, increments a Prometheus error counter, and returns a zero/neutral result.

See Chapter for full documentation of each signal.

Attacker Attribution (Layer 11)

Layer 11 correlates JA4, JA4X, and JA4T fingerprints across IPs observed by the analytics node. It looks for:

- Multiple IPs sharing the same JA4/JA4T profile (coordinated scanning)
- JA4X extension ordering inconsistencies with claimed browser identity
- Fingerprint drift — the same IP changing fingerprints across connections

Attribution findings are written by the analytics node to `findings:{ip}` in Redis (TTL=300s) and read back by the pipeline at layer 11.

Behavioural Analysis (Layer 12)

Layer 12 detects behavioural patterns that individual connections cannot reveal:

- **Sequential probing** — connections to incrementally varying SNI values from the same source IP, characteristic of host enumeration
- **Coordinated bursts** — sudden increases in connection rate from multiple IPs sharing a fingerprint (botnet activation pattern)
- **Fingerprint drift** — the same IP cycling through multiple JA4 values to evade per-fingerprint rate limits

Risk Scorer (Layer 13)

Score Aggregation Algorithm

The risk scorer aggregates all `RiskSignal` objects from layers 3–12 into a single integer score in the range 0–100. The aggregation algorithm is confidence-weighted:

1. Each signal contributes `signal.score × signal.weight` to the total
2. Weights are normalised to sum to 1.0 across all active signals
3. The raw weighted sum is clamped to [0, 100]
4. Signal scores from external services that are unreachable contribute 0 (fail open — a missing signal never increases the score)

Score Semantics

Score	Interpretation	Typical source	Default action at <code>fail=100</code>
0–19	Clean	Browser traffic, known-good fingerprints	Allow
20–39	Low risk	Datacenter IP with clean fingerprint	Allow or flag
40–59	Moderate risk	Scanner, unresolved PTR, high-confidence AbuseIPDB	Rate limit
60–79	High risk	Tor exit, known-bad ASN, beaconing pattern	Tarpit
80–100	Critical	Confirmed C2 fingerprint, Spamhaus signal, banned IP	Block or ban

Table 6: Score ranges and their interpretation

Action Decider (Layer 14)

The Dial

The `dial` value (0–100) controls how aggressively scores are translated into blocking actions. At `dial=0`, every connection is allowed regardless of score — the system is in pure monitor mode. At `dial=100`, the thresholds configured in `security.rate_limit_strategies` apply in full.

Dial Increment Policy

The dial should be raised incrementally. A jump from 0 to 100 without first validating the score distribution risks generating false positives at scale. The recommended approach is to raise the dial to 20 and observe for 24 hours, then to 50, then to 100.

The Six Actions

Action	Badge	Connection fate	Notes	Table 7: Action definitions and connection handling behaviour
allow		Forwarded to backend unchanged	ClientHello bytes replayed to backend; proxy becomes transparent relay	
flag		Forwarded to backend	Connection allowed but high-priority event emitted to analytics stream	
rate_limit		Queued or delayed	Token bucket or sliding window rate limiting applied per-IP or per-JA4	
tarpit		Artificially slowed	Data delivered to backend at throttled rate; consumes attacker connection slots	
block		TCP RST sent	Connection closed immediately; no data forwarded; TTL-free (not persistent)	
ban		TCP RST sent	IP added to Redis ban SET with TTL; all future connections blocked immediately at layer 0	

Ban Escalation

Ban escalation applies when an IP continues to trigger blocks after an initial ban expires. The ban TTL doubles on each re-ban up to the configured maximum:

- First ban: `ban_duration` seconds (default 300)
- Second ban: `ban_duration` × 2 seconds
- Subsequent bans: capped at 604800 seconds (7 days)

Ban state is stored in Redis as `ban:{ip}` with the TTL set at write time. The proxy reads ban state from its local in-process cache first (TTL=30s) to avoid Redis round-trips for known-banned IPs.

Multi-Strategy Policy

Three rate-limiting strategies run simultaneously, each tracking a different key:

by_ip Rate window keyed by client IP. Detects high-volume floods from a single address.

by_ja4 Rate window keyed by JA4 fingerprint. Detects distributed scanning sharing a tool fingerprint.

by_ip_ja4_pair Rate window keyed by (IP, JA4) pair. Detects IP-rotation evasion where each IP presents the same fingerprint.

The `security.multi_strategy_policy` setting controls how strategy results are combined:

any Block if *any* strategy triggers (most aggressive)

majority Block if *two or more* strategies trigger (default)

all Block only if *all* strategies trigger (most conservative)

Pipeline Performance Characteristics

Zone	Python (asyncio)	Go	Notes
Fast-path bypass (cache hit)	<100 μ s	<10 μ s	Local LRU cache, no Redis
Fast-path bypass (Redis)	200–400 μ s	150–300 μ s	Single Redis GET
Signal collection (parallel)	1–5 ms	0.1–1 ms	External services: AbuseIPDB 50ms (cached)
Scoring and decision	<100 μ s	<10 μ s	Pure computation
Total (cached, fast path)	<500 μ s	<50 μ s	Typical cached connection
Total (full pipeline, no cache)	2–10 ms	0.3–2 ms	First-seen IP, all signals

Table 8: Expected latency contribution by pipeline zone

AbuseIPDB Latency

AbuseIPDB lookups use aggressive caching (TTL=3600s by default). A cache hit adds zero latency. Only cache misses (first-seen IP) incur the network round-trip, which is 50ms and runs asynchronously — it does not block the connection decision.

Configuration Reference

Every behavioural parameter in JA4proxy is configurable in `config/proxy.yml`. All keys except `listen port`, `Redis URL`, and `TLS certificate paths` support hot-reload via `SIGHUP` or the Management UI.

THE CONFIGURATION FILE is `config/proxy.yml`. Environment variable interpolation is supported: `${VAR_NAME}` is replaced with the environment variable at startup. Secrets (API keys, Redis password) must be provided via environment variables — never hard-coded in the config file.

Hot Reload Behaviour

On `SIGHUP`, JA4proxy re-reads `config/proxy.yml` and applies all changes that support hot-reload. A `config_reload` event is emitted to the policy audit log in Redis. The Management UI sends a Redis `pub/sub` message to trigger reload without a Unix signal.

Section	Hot-reloadable	Notes	Table 9: Hot-reload support by configuration section
<code>proxy.bind_host / bind_port</code>	No	Requires process restart	
<code>proxy.backend_host / backend_port</code>	Yes	Applies to new connections	
<code>dial</code>	Yes	Applies immediately to all new decisions	
<code>security_policy</code>	Yes	New bypass states apply to next connection	
<code>geoip</code>	Yes	New country lists apply immediately	
<code>security (lists)</code>	Yes	List changes apply to next connection	
<code>abuseipdb</code>	Yes	Cache TTL change affects new lookups	
<code>rdap</code>	Yes	<code>block_expansion toggle</code> applies immediately	
<code>spamhaus</code>	No (<code>refresh_interval</code>)	Next trie refresh uses new URL	
<code>decision_cache</code>	Yes	New TTLs apply to decisions cached after reload	
<code>redis.url / redis.password</code>	No	Requires process restart	

proxy — Core Listener Settings

```
1 proxy:
```

Key	Type	Default	Description
bind_host	string	"0.0.0.0"	Listen address for incoming TCP connections from HAProxy
bind_port	integer	8080	Listen port. Must match HAProxy backend configuration
backend_host	string	<i>required</i>	Hostname or IP of the protected backend. Must be set
backend_port	integer	443	Backend port. Typically 443 for HTTPS

Table 10: proxy section — core listener settings

```

2 bind_host: "0.0.0.0"
3 bind_port: 8080
4 backend_host: "backend" # Docker service name or hostname
5 backend_port: 443

```

dial — Blocking Aggression

Key	Type	Default	Description
dial	integer	0	Blocking aggression: 0 = monitor only (no blocking); 100 = full blocking per configured thresholds. Intermediate values scale blocking proportionally. Raise incrementally.

Table 11: dial setting

Never Deploy with dial=100 on First Install

Always start at dial: 0. Observe the score distribution for at least 24 hours before raising the dial. A premature jump to 100 can block legitimate traffic before you have validated your threshold configuration.

security_policy — Bypass Conditions

Each bypass condition has a single enabled boolean key. All default to true.

```

1 security_policy:
2   alpn_browser_bypass:
3     enabled: true # h2/h1 ALPN -> ALLOW without scoring
4   ja4_whitelist_bypass:
5     enabled: true # JA4 in whitelist -> ALLOW without scoring
6   mtls_bypass:
7     enabled: true # Valid mTLS cert -> ALLOW without scoring
8   static_ip_allowlist:
9     enabled: true # IP in static allowlist -> ALLOW without
10    scoring
11   ja4_blacklist_bypass:
12     enabled: true # JA4 in blacklist -> immediate RST, no
13    scoring
14   country_blacklist_bypass:

```

Key	Default	Effect when true	Risk if disabled
alpn_browser_bypass.enabled	true	h2/h1 ALPN → AL-LOW without scoring	Browser traffic scored; false positive risk elevated
ja4_whitelist_bypass.enabled	true	JA4 in whitelist → AL-LOW without scoring	Whitelisted fingerprints scored; may be blocked if above threshold
mtls_bypass.enabled	true	Valid mTLS cert → AL-LOW without scoring	mTLS clients scored; may be blocked if above threshold
static_ip_allowlist.enabled	true	IP in allowlist → AL-LOW without scoring	Allowlisted IPs scored
ja4_blacklist_bypass.enabled	true	JA4 in blacklist → TCP RST immediately	Blacklisted fingerprints only blocked if score exceeds dial threshold
country_blacklist_bypass.enabled	true	Country in blacklist → BLOCK immediately	Blacklisted countries only blocked by scorer
spamhaus_bypass.enabled	true	Spamhaus DROP match → BLOCK immediately	Spamhaus matches produce RiskSignal(+80) instead of hard block
tls_version_bypass.enabled	true	TLS 1.0/1.1/SSLv3 → TCP RST immediately	Old TLS versions produce risk score contribution instead of hard reject

```

13     enabled: true # Country in GeoIP blacklist -> immediate
    block
14 spamhaus_bypass:
15     enabled: true # Spamhaus DROP/EDROP -> immediate block
16 tls_version_bypass:
17     enabled: true # TLS 1.0/1.1/SSLv3 -> immediate RST

```

geoip — Geographic Controls

Key	Type	Default	Description
country_whitelist_enabled	boolean	true	Enable country allowlist mode. If true, only countries in country_whitelist are allowed
country_whitelist	list	["IE", "GB", "IM", "US"]	ISO 3166-1 alpha-2 country codes permitted when whitelist mode is active
country_blacklist_enabled	boolean	true	Enable country blacklist mode. Listed countries are immediately blocked
country_blacklist	list	["KP", "RU"]	ISO 3166-1 alpha-2 country codes to block immediately

Whitelist and Blacklist Together

Both modes can be active simultaneously. The whitelist check runs first. A country not on the whitelist (when whitelist mode is active) is blocked independently of whether it appears on the blacklist. Use whitelist mode for strict geographic restric-

tions (allowlist your expected countries), and blacklist mode for blocking specific known-bad countries when you serve a broad geographic audience.

```

1 geoip:
2   country_whitelist_enabled: true
3   country_whitelist: ["IE", "GB", "IM", "US"]
4   country_blacklist_enabled: true
5   country_blacklist: ["KP", "RU"]

```

security — Fingerprint Lists and Rate Limits

Fingerprint Lists

Key	Type	Default	Description	Table 14: security section — fingerprint lists
whitelist	list of strings	[]	Exact JA4 fingerprints to whitelist. Loaded into Redis <code>ja4:whitelist</code> and an in-process set	
whitelist_patterns	list of strings	["h2", "h1"]	ALPN patterns that trigger the ALPN browser bypass. Separate from the JA4 whitelist	
blacklist	list of strings	[]	Exact JA4 fingerprints to blacklist. Loaded into Redis <code>ja4:blacklist</code>	
fingerprint_labels	map	{}	Human-readable names for fingerprints. Used in logs and metrics	
multi_strategy_policy	string	"majority"	How to combine rate limit strategy results: any / majority / all	

Rate Limit Strategies

Three rate-limiting strategies are defined under `security.rate_limit_strategies`. Each has an identical structure:

Key	Type	Default	Description	Table 15: rate_limit_strategies — per-strategy keys
thresholds.suspicious	integer	—	Connections/second at which the strategy flags as suspicious	
thresholds.block	integer	—	Connections/second at which the strategy triggers a block	
thresholds.ban	integer	—	Connections/second at which the strategy triggers a permanent ban	
action	string	—	Action to take when threshold is crossed: tarpit or block	
ban_duration	integer	300	Initial ban TTL in seconds. Escalates on re-ban	

```

1 security:
2   whitelist:

```

```

3   - "t13d1516h2_8daaf6152771_02713d6af862" # Chrome 120+
4   whitelist_patterns:
5     - "h2"
6     - "h1"
7   blacklist:
8     - "t13d190900_9dc949149365_97f8aa674fd9" # Sliver C2
9   fingerprint_labels:
10    "t13d1516h2_8daaf6152771_02713d6af862": "Chrome 120+"
11   multi_strategy_policy: "majority"
12   rate_limit_strategies:
13     by_ip:
14       thresholds: {suspicious: 50, block: 200, ban: 500}
15       action: "tarpit"
16       ban_duration: 300
17     by_ja4:
18       thresholds: {suspicious: 20, block: 100, ban: 200}
19       action: "block"
20       ban_duration: 300
21     by_ip_ja4_pair:
22       thresholds: {suspicious: 20, block: 50, ban: 100}
23       action: "tarpit"
24       ban_duration: 300

```

abuseipdb — Reputation Lookup

Key	Type	Default	Description	Table 16: abuseipdb section
enabled	boolean	true	Enable AbuseIPDB lookups	
api_key	string	<i>required</i>	AbuseIPDB API key. Must be an env var reference: <code>\${ABUSEIPDB_API_KEY}</code>	
cache_ttl	integer	3600	Seconds to cache AbuseIPDB responses in Redis	
min_score_to_signal	integer	50	Minimum AbuseIPDB score (0–100) that produces a RiskSignal. Never hard-block below 50	

AbuseIPDB Score 50 Floor

The `min_score_to_signal` floor exists because AbuseIPDB scores below 50 are common for shared infrastructure (NAT gateways, VPNs, corporate egress). Hard-blocking based on scores below 50 has an unacceptably high false positive rate. The default of 50 should not be lowered without evaluating the false positive impact on your traffic.

```

1 abuseipdb:
2   enabled: true
3   api_key: "${ABUSEIPDB_API_KEY}"
4   cache_ttl: 3600
5   min_score_to_signal: 50

```

rdap — WHOIS/RDAP Org Reputation

Key	Type	Default	Description
enabled	boolean	true	Enable RDAP/WHOIS enrichment lookups
block_expansion	boolean	false	Automatically block the entire /24 (IPv4) or /48 (IPv6) of a blocked IP. Must be explicitly enabled
max_expansion	integer	24	Maximum prefix length for block expansion. Never set below 24 for IPv4

Table 17: rdap section

Block Expansion Off by Default

block_expansion: true will cause JA4proxy to block the entire /24 subnet when a single IP is blocked. On residential ISPs and shared hosting, an entire /24 may serve thousands of unrelated users. Enable only after confirming that the blocked IPs are from dedicated attacker infrastructure with no residential or business customers.

```

1 rdap:
2   enabled: true
3   block_expansion: false      # secops must explicitly opt in
4   max_expansion: 24          # never beyond /24 IPv4, /48 IPv6

```

spamhaus — DROP/EDROP Blocklists

Key	Type	Default	Description
drop_list_url	string	https://www.spamhaus.org/drop/drop.txt	URL to the Spamhaus DROP list
edrop_list_url	string	https://www.spamhaus.org/drop/edrop.txt	URL to the Spamhaus EDROP list
refresh_interval	integer	3600	Seconds between trie refresh cycles

Table 18: spamhaus section

```

1 spamhaus:
2   drop_list_url: "https://www.spamhaus.org/drop/drop.txt"
3   edrop_list_url: "https://www.spamhaus.org/drop/edrop.txt"
4   refresh_interval: 3600

```

redis — Connection Settings

Key	Type	Default	Description
url	string	"redis://redis:6379"	Redis connection URL. For Unix socket: unix:///var/run/redis/redis.sock
password	string	optional	Redis AUTH password. Use env var: \${REDIS_PASSWORD}

Table 19: redis section

Unix Domain Socket Performance

For deployments where JA4proxy and Redis are on the same host, using a Unix domain socket for the Redis URL reduces latency by approximately 30–40% compared to TCP loopback. Set `url: "unix:///var/run/redis/redis.sock"` and ensure the socket path is accessible inside the container.

```
1 redis:
2   url: "redis://redis:6379"
3   password: "${REDIS_PASSWORD}"
```

decision_cache — Per-Client Decision Cache

The in-process decision cache lets JA4proxy skip the full signal pipeline for a recently-seen client. Per ADR-003 it is keyed **per client** — the client IP combined with its JA4 fingerprint — so a decision made for one client is never served to another that merely shares a TLS fingerprint. Entries use **asymmetric TTLs**: an `allow` is cached for a long window (a client judged legitimate stays fast without re-scoring), while every blocking action (`block`, `ban`, `tarpit`, `rate_limit`, `flag`) is cached only briefly so a mistaken or stale block self-heals and a lifted ban takes effect almost immediately.

Key	Type	Default	Description
<code>allow_ttl_seconds</code>	integer	1800	TTL for cached <code>allow</code> decisions (long window; ADR-003)
<code>block_ttl_seconds</code>	integer	30	TTL for cached blocking actions (short window; self-healing)
<code>max_entries</code>	integer	10000	Cap on cached decisions; bounds memory

Table 20: `decision_cache` section

Both TTLs are hot-reloadable. Zero or absent values fall back to the defaults above, so a partial configuration can never disable expiry.

```
1 decision_cache:
2   allow_ttl_seconds: 1800
3   block_ttl_seconds: 30
4   max_entries: 10000
```

Complete Configuration Example

```
1 # config/proxy.yml - complete reference configuration
2
3 proxy:
4   bind_host: "0.0.0.0"
5   bind_port: 8080
6   backend_host: "backend"
7   backend_port: 443
```

```

8
9 # 0 = monitor only. Raise incrementally after validating score
   distribution.
10 dial: 0
11
12 security_policy:
13   alpn_browser_bypass:
14     enabled: true
15   ja4_whitelist_bypass:
16     enabled: true
17   mtls_bypass:
18     enabled: true
19   static_ip_allowlist:
20     enabled: true
21   ja4_blacklist_bypass:
22     enabled: true
23   country_blacklist_bypass:
24     enabled: true
25   spamhaus_bypass:
26     enabled: true
27   tls_version_bypass:
28     enabled: true
29
30 geoip:
31   country_whitelist_enabled: true
32   country_whitelist: ["IE", "GB", "IM", "US"]
33   country_blacklist_enabled: true
34   country_blacklist: ["KP", "RU"]
35
36 security:
37   whitelist:
38     - "t13d1516h2_8daaf6152771_02713d6af862" # Chrome 120+
39   whitelist_patterns:
40     - "h2"
41     - "h1"
42   blacklist:
43     - "t13d190900_9dc949149365_97f8aa674fd9" # Sliver C2
44   fingerprint_labels:
45     "t13d1516h2_8daaf6152771_02713d6af862": "Chrome 120+"
46   multi_strategy_policy: "majority"
47   rate_limit_strategies:
48     by_ip:
49       thresholds: {suspicious: 50, block: 200, ban: 500}
50       action: "tarpit"
51       ban_duration: 300
52     by_ja4:
53       thresholds: {suspicious: 20, block: 100, ban: 200}
54       action: "block"
55       ban_duration: 300
56     by_ip_ja4_pair:
57       thresholds: {suspicious: 20, block: 50, ban: 100}
58       action: "tarpit"
59       ban_duration: 300
60
61 abuseipdb:
62   enabled: true
63   api_key: "${ABUSEIPDB_API_KEY}"

```



```
64  cache_ttl: 3600
65  min_score_to_signal: 50
66
67  rdap:
68    enabled: true
69    block_expansion: false
70    max_expansion: 24
71
72  spamhaus:
73    drop_list_url: "https://www.spamhaus.org/drop/drop.txt"
74    edrop_list_url: "https://www.spamhaus.org/drop/edrop.txt"
75    refresh_interval: 3600
76
77  redis:
78    url: "redis://redis:6379"
79    password: "${REDIS_PASSWORD}"
```


Signal Reference

A signal is a scored observation. Each of the eight signal checks contributes a score in $[0, 100]$ and a confidence weight. The risk scorer combines them. No single signal can block a connection — only the combined score, acting through the dial, produces blocking actions.

ALL EIGHT SIGNALS RUN CONCURRENTLY after the fast-path bypass zone. Each signal is independent. An unreachable external service contributes zero — it never increases the score (fail open). Signal results are logged in the `signals` field of every connection event.

Signal Score Model

Each signal produces a `RiskSignal` with three fields:

score Integer 0–100. Higher means more suspicious.

weight Float 0.0–1.0. The confidence weight of this signal type. High-confidence signals (TLS version, JA4 blacklist category) have higher weights. Low-certainty signals (RDAP org, passive DNS) have lower weights.

reason Human-readable explanation for the log.

The risk scorer computes: $\text{final_score} = \text{clamp}(\sum_i \text{score}_i \times \text{weight}_i, 0, 100)$ where weights are normalised to sum to 1.0 across all active signals.

Table 21: Signal summary — all eight signals					
#	Signal	Phase	Default weight	External service	Failure behaviour
1	TLS version & cipher	3	0.20	None	N/A — local check
2	SNI analysis	4	0.15	None	N/A — local check
3	TCP & connection behaviour	5	0.15	Redis	Score=0 on Redis error
4	ASN classification	6	0.15	MaxMind DB (mmap)	Score=0 if DB unavailable
5	FCrDNS enrichment	7	0.10	DNS resolver	Score=0 on DNS error
6	Beaconing detector	9	0.10	Redis	Score=0 on Redis error
7	AbuseIPDB score	10	0.10	AbuseIPDB API	Score=0 on API error
8	RDAP org reputation	11	0.05	RDAP/WHOIS APIs	Score=0 on API error

Signal 1 — TLS Version and Cipher Check

Phase: 3 **Module:** `src/security/tls_enforcer.py`

What It Measures

This signal checks the TLS protocol version and cipher suite list from the ClientHello against a set of hard rules. TLS 1.0 and 1.1 are deprecated and have known vulnerabilities; SSLv3 is critically broken. Weak cipher suites indicate either an outdated client or deliberate downgrade probing.

Score Computation

When `security_policy.tls_version_bypass.enabled` is true, TLS 1.0/1.1/SSLv3 trigger an immediate TCP RST (before scoring). When the bypass is disabled, they contribute a RiskSignal instead:

Condition	Score	Notes
TLS 1.3, strong ciphers	0	Clean
TLS 1.2, ECDHE + AES-GCM / ChaCha20	0	Acceptable
TLS 1.2 with weak ciphers (RC4, DES, 3DES, NULL, EXPORT)	40	Routed through scorer
TLS 1.1 (bypass disabled)	70	Deprecated protocol
TLS 1.0 (bypass disabled)	80	Deprecated protocol
SSLv3 (bypass disabled)	95	Critically deprecated

Table 22: TLS version score contributions

Configuration Keys

`security_policy.tls_version_bypass.enabled` — when true, old TLS versions trigger TCP RST directly. When false, they contribute a risk score.

Failure Behaviour

This signal performs only local parsing of the ClientHello. There is no external service dependency. It cannot fail.

Signal 2 — SNI Analysis

Phase: 4 **Module:** `src/security/sni_analyzer.py`

What It Measures

Server Name Indication (SNI) is an extension in the ClientHello that specifies which hostname the client is trying to reach. Legitimate browsers always include SNI when connecting to HTTPS services. Missing SNI, IP-literal SNI, and algorithmically generated domain names are all strong signals of automated tools or malware.

Score Computation

DGA Detection Algorithm

Domain Generation Algorithm detection uses Shannon entropy of the domain label characters. A real domain like `example.com` has low

Condition	Score	Notes	Table 23: SNI signal score contributions
SNI present, matches expected hostname	0	Normal browser or API client	
SNI present, unexpected hostname	10–30	May be misconfigured client or proxy	
SNI absent	50	Scanners and tools typically omit SNI	
SNI is an IP literal (e.g., 192.168.1.1)	60	Invalid SNI; tools and scanners	
SNI is DGA-like (high entropy, no real TLD)	40–80	Score based on entropy measurement	

entropy; a DGA domain like `xk9p2mq4vr1.cn` has high entropy. The threshold is tunable:

- Entropy < 3.5 bits/character: clean
- Entropy 3.5–4.2 bits/character: moderate score (40)
- Entropy > 4.2 bits/character: high score (80)

Additionally, the TLD is checked against a list of TLDs commonly used in DGA campaigns. A domain matching a flagged TLD with high entropy scores at the upper end of the range.

Failure Behaviour

This signal performs only local computation. There is no external service dependency. It cannot fail.

Signal 3 — TCP and Connection Behaviour

Phase: 5 **Modules:** `src/security/tcp_analyzer.py`, `src/security/mtls.py`

What It Measures

TCP and connection behaviour signals capture how a client connects over time, not just what it sends. Automated tools have different connection patterns from real users: they connect more frequently, resume sessions less often, and maintain more simultaneous connections.

JA4T Fingerprinting

JA4T is the TLS transport fingerprint — a hash of the TCP window size, TCP options (MSS, window scale, SACK, timestamps), and their ordering. Unlike JA4 (which fingerprints the TLS application layer), JA4T fingerprints the TCP/IP stack implementation. Together, JA4 + JA4T can distinguish a real Chrome browser from a tool that has copied Chrome's JA4 value but runs on a different OS or with a modified TCP stack.

Score Components

Redis Keys Used

- `sess:{ip}` — Hash of total and resumed session counts
- `conn:{ip}` — Current concurrent connection count (INCR/DECR)

Component	Score contribution	Notes	Table 24: TCP behaviour signal score components
JA4T mismatch with claimed browser	+20	JA4T profile inconsistent with JA4 fingerprint label	
Session resumption ratio < 10%	+15	Legitimate browsers resume sessions aggressively	
Concurrent connections > 20	+30	Legitimate browsers maintain 6–8 connections	
Very short connection lifespan (< 0.1s)	+20	Scanner pattern — connect, sniff, disconnect	
First-seen IP (no return visit history)	+5	Weak signal; most IPs start here	
Return visitor with all blocked history	+25	IP that only ever got blocked	

- `visit:{ip}` — Hash of `first_seen`, `total`, `allowed`, `blocked`

Failure Behaviour

If Redis is unreachable, all Redis-backed sub-signals return `score=0`. The signal as a whole returns `score=0`. The proxy logs the Redis error and increments `ja4proxy_redis_operations_total` with `result="error"`.

Signal 4 — ASN Classification

Phase: 6 **Module:** `src/security/asn_classifier.py`

What It Measures

The Autonomous System Number (ASN) associated with the client IP identifies the network organisation that owns the IP block. Legitimate browser traffic overwhelmingly originates from residential ISPs, mobile carriers, and corporate networks. Automated scanners and bots are disproportionately hosted in datacenters and cloud providers.

Score Computation

ASN classification uses:

1. The MaxMind GeoLite2 ASN database (memory-mapped for zero-copy lookup)
2. A curated datacenter ASN list in `config/asn_datacenter_list.yml`
3. The Tor exit node list (fetched periodically from `check.torproject.org`)

Failure Behaviour

If the MaxMind database file is missing or corrupt, this signal returns `score=0`. The proxy logs the failure (there is no dedicated error metric for a missing GeoIP database; `ja4proxy_signal_latency_seconds` records only timing). The GeoLite2 database should be updated monthly and validated at startup.

ASN category	Score	Notes	Table 25: ASN signal score contributions
Residential ISP	0	Expected for browser traffic	
Mobile carrier	0	Expected for mobile browser traffic	
Corporate network	5	Minor flag — APIs often originate here	
Hosting / datacenter (general)	25	Common for cloud-hosted scanners	
Known VPN provider	30	Often used to evade geo-blocking	
Tor exit node	65	Significant signal — anonymisation tool	
Known botnet/abuse ASN	80	From curated bad-org list	

Signal 5 — FCrDNS Enrichment

Phase: 7 **Module:** `src/security/dns_enrichment.py`

What It Measures

Forward-confirmed reverse DNS (FCrDNS) validates that the PTR record for a client IP resolves to a hostname, and that hostname's A/AAAA record resolves back to the same IP. Residential ISPs, cloud providers, and CDNs all maintain consistent FCrDNS. Ephemeral scanner infrastructure typically does not.

Classification Logic

Classification	Score	Notes	Table 26: FCrDNS signal classification and scores
FCrDNS valid, residential pattern	0	PTR resolves to ISP hostname pattern	
FCrDNS valid, cloud/datacenter	10	e.g. <code>ec2-*.compute.amazonaws.com</code>	
PTR absent or NXDOMAIN	30	No reverse DNS — common for scanners	
PTR present but FCrDNS fails	35	Inconsistent DNS — suspicious	
PTR matches known scanning hostname	60	e.g. Shodan, Censys, known scanner infra	

Async Lookup Architecture

DNS lookups are fire-and-forget: `asyncio.create_task(dns_lookup(ip))` is called at pipeline start. Results are written to a short-lived in-process cache. If the lookup has not completed when the scorer runs, the signal returns `score=0` for this connection (fail open). Results are available for subsequent connections from the same IP.

Failure Behaviour

DNS resolution errors (NXDOMAIN, SERVFAIL, timeout) cause this signal to return `score=0`. The failure is logged at DEBUG level (DNS failures are common and non-actionable).

Signal 6 — Beaconsing Detector

Phase: 9 **Module:** `src/security/beaconsing_detector.py`

What It Measures

Beaconing is the pattern of regular, low-jitter connections from the same source — characteristic of C2 implants and automated scheduled tasks rather than human browsing. Human browsing has highly variable inter-arrival times (IAT). Beaconing tools produce IAT distributions with low coefficient of variation.

IAT Algorithm

1. Connection timestamps are stored in Redis Sorted Set `beacon:{ip}:{ja4}` (score = Unix timestamp in milliseconds)
2. The sliding window retains the last N timestamps within a 30-minute window
3. **Inter-arrival times** are computed as consecutive differences: $IAT_i = t_{i+1} - t_i$
4. The **coefficient of variation** (CV) is computed: $CV = \sigma / \mu$
5. Low CV indicates regular beaconing; high CV indicates human browsing

CV range	Score	Notes
< 5 connections in window	0	Insufficient data; no score assigned
$CV > 0.8$	0	High variability — human pattern
$CV\ 0.4\text{--}0.8$	20	Moderate regularity
$CV\ 0.1\text{--}0.4$	50	Strong beaconing pattern
$CV < 0.1$	75	Near-perfect regularity — almost certainly automated

Table 27: Beaconing signal score thresholds

Redis Keys Used

`beacon:{ip}:{ja4}` — Sorted Set where member = timestamp and score = timestamp. `ZRANGEBYSCORE` is used to retrieve the sliding window. `ZADD` appends new timestamps. Old members (outside the window) are removed with `ZREMRANGEBYSCORE`.

Failure Behaviour

If Redis is unreachable, this signal returns `score=0`. Redis writes are fire-and-forget and do not block the pipeline.

Signal 7 — AbuseIPDB Score

Phase: 10 **Module:** `src/security/abuseipdb.py`

What It Measures

AbuseIPDB is a community-maintained database of IP addresses reported for malicious activity (scanning, brute-force, spam, C2 traffic). Each IP has a confidence score (0–100) reflecting how frequently it has been reported and how recently.

Score Computation

AbuseIPDB confidence	Risk signal score	Notes	Table 28: AbuseIPDB signal score mapping
0–49	0	Below minimum threshold; no signal	
50–74	30	Moderate abuse history	
75–89	60	Significant abuse history	
90–100	80	Very high confidence abuse	
API error / timeout	0	Fail open — never increases score	

Caching Architecture

AbuseIPDB responses are cached in Redis as `abuseipdb:{ip}` (JSON, TTL=3600s by default). Cache hits return immediately with zero API call. Cache misses trigger an async API call (fire-and-forget) and return `score=0` for the current connection. The result is available for subsequent connections from the same IP.

This design means the *first* connection from a never-seen-before IP pays no AbuseIPDB penalty — it is allowed while the lookup runs. This is intentional (fail open).

Failure Behaviour

API errors, HTTP 429 rate limit responses, and network timeouts all produce `score=0`. The error is logged with context and the `ja4proxy_abuseipdb_lookups_total{result="error"}` counter is incremented. The proxy continues functioning with zero AbuseIPDB signal.

Signal 8 — RDAP Org Reputation

Phase: 11 **Module:** `src/security/rdap_enrichment.py`

What It Measures

RDAP (Registration Data Access Protocol) and WHOIS provide the organisation that owns an IP block. JA4proxy maintains a curated list of known-bad organisations in `config/known_bad_orgs.yml`. Matching the RDAP organisation name against this list provides a signal about whether the client IP is associated with organisations that systematically operate malicious infrastructure.

Score Computation

Condition	Score	Notes	Table 29: RDAP signal score contributions
Known-good org (major cloud, CDN)	0	AWS, GCP, Azure, Cloudflare, Akamai	
Unknown org	5	Default for unrecognised registrants	
Org name matches known-bad list	60–80	Score from <code>known_bad_orgs.yml</code>	
RDAP lookup failed	0	Fail open	

Block Expansion

When `rdap.block_expansion: true` (disabled by default), blocking an IP from a known-bad org triggers an automatic CIDR block of the entire /24 (IPv4) or /48 (IPv6) prefix containing that IP. The expanded block is written to Redis and picked up by the Layer 3 CIDR trie refresh.

Block Expansion Risk

Block expansion must never be enabled without manual review of the affected prefix. RDAP data is sometimes stale, incorrect, or shared between legitimate and malicious operators. A rogue /24 expansion affecting a major ISP block can lock out thousands of legitimate users.

WHOIS Pivot

The RDAP enrichment module also performs WHOIS pivoting: it looks up the registrant email domain for the IP block's registrant record. A registrant email from a known throwaway or bulletproof hosting domain adds an additional score contribution.

Failure Behaviour

RDAP API errors, timeouts, and malformed responses all return `score=0`. RDAP queries are cached in Redis (`rdap:{ip_or_prefix}`, `TTL=86400s`) to avoid repeated lookups for the same prefix.

Phase 203 Signals (Go Production Runtime)

PHASE 203 CLOSES FIVE PRODUCTION-CRITICAL SIGNAL GAPS IN THE GO PROXY. These signals run *in addition* to the eight described above; they are documented here as a list because the full algorithmic detail is maintained in the phase document (`docs/phases/PHASE_203.md`) rather than reproduced in this reference manual. Each signal is independently configurable and follows the same fail-open contract as Signals 1–8.

TAP OS-mismatch (203a) Reads JA4T fingerprints written to Redis by the Phase 20 TAP node (key `fp:os:ip:{ip}`). When the JA4-claimed OS-class disagrees with the TAP-observed OS-class, a signal is emitted. The inline Go proxy cannot compute JA4T itself — by the time `accept()` returns, the kernel has consumed the SYN. If the TAP node is not deployed, the lookup returns nil and the signal is silently dormant (fail open). See `docs/phases/PHASE_203.md §203a` for the architectural rationale.

JA4 TLS-version mismatch (203b) Catches TLS-version spoofing where the JA4 prefix (`t13/t12/...`) disagrees with the TLS version actually

negotiated. A common evasion pattern for tools that hand-craft a ClientHello matching a modern browser but cannot complete a real TLS 1.3 handshake.

Weak-cipher parity (203c) The Go `weakCipherSet` is now synchronised to *exactly* the 40 cipher suites enumerated in Python's `WEAK_CIPHERS`. A drift between the two implementations had previously let some weak suites score zero in Go that scored 40 in Python. Verified by a parity test against the Python oracle.

DGA parity (203d) The Go `dgaConfidence()` algorithm has been reported to match Python's `dga_score()` rule-for-rule. The Shannon-entropy thresholds, TLD list, and cap-out scoring now match across implementations.

Deep health (203e) `/health/deep` adds component checks for tarpit saturation and (optionally) GeoIP database freshness, with anti-flap hysteresis so transient blips do not flap the endpoint. `/health` (the k8s liveness probe) is deliberately left untouched and remains a tight TCP-level reachability check.

Phase 203 signals are Go-only

These five signals exist in the Go production runtime. The Python prototype contains the original implementations from which 203c (weak ciphers) and 203d (DGA) were ported; 203a/203b/203e are net-new in the Go runtime and have no Python counterpart.

Metrics Reference

All JA4proxy metrics are prefixed `ja4proxy_`. They are exposed on the `/metrics` HTTP endpoint (default port 9090) in Prometheus text format and scraped by the Prometheus instance at `:9090`.

METRIC NAMING FOLLOWS the Prometheus naming convention: `ja4proxy_{subsystem}_{metric_name}_{unit}`. Counters always end in `_total`. Histograms emit `_bucket`, `_count`, and `_sum` automatically. All metrics are registered at startup; a metric that has never been incremented will still appear in `/metrics` with a zero value.

Connection Metrics

`ja4proxy_connections_total`

Field	Value	Table 30: ja4proxy_connections_total — connection outcome counter
Type	Counter	
Labels	action: allow block tarpit ban flag rate_limit	
Description	Total connections processed, labelled by the final action taken	

Example PromQL:

```
1 # Current connection rate (all outcomes)
2 rate(ja4proxy_connections_total[5m])
3
4 # Block ratio
5 rate(ja4proxy_connections_total{action="block"}[5m])
6 / rate(ja4proxy_connections_total[5m])
7
8 # Breakdown by action
9 sum by (action) (rate(ja4proxy_connections_total[1m]))
```

`ja4proxy_connections_active`

Type	Gauge	Table 31: ja4proxy_connections_active — current active connections
Labels	None	gauge
Description	Number of connections currently in the pipeline or forwarding phase	

Bypass Metrics

ja4proxy_bypass_total

Type	Counter	Table 32: ja4proxy_bypass_total — fast-path bypass counter by bypass type
Labels	type: alpn ja4_whitelist mtls static_ip ja4_blacklist country spamhaus tls_version	
Description	Total connections short-circuited by each bypass condition	

Monitor Bypass Volumes

Monitor `ja4proxy_bypass_total` when modifying bypass configuration. A sudden drop in `alpn` bypass traffic may indicate HAProxy misconfiguration or an upstream change in ALPN negotiation.

Example PromQL:

```

1 # ALPN bypass rate
2 rate(ja4proxy_bypass_total{type="alpn"}[5m])
3
4 # Spamhaus blocks per minute
5 rate(ja4proxy_bypass_total{type="spamhaus"}[1m]) * 60

```

Risk Scoring Metrics

ja4proxy_risk_score

Type	Histogram	Table 33: ja4proxy_risk_score — score distribution histogram
Labels	None	
Buckets	0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100	
Description	Distribution of computed risk scores across all fully scored connections (bypassed connections are excluded)	

This histogram is the primary tool for evaluating threshold configuration before raising the dial. Plot the distribution in Grafana and select a threshold that captures known-bad traffic with minimal false positive exposure.

Example PromQL:

```

1 # 95th percentile score
2 histogram_quantile(0.95, rate(ja4proxy_risk_score_bucket[5m]))
3
4 # Fraction of connections scoring above 60
5 (
6   sum(rate(ja4proxy_risk_score_bucket{le="100"}[5m]))
7   - sum(rate(ja4proxy_risk_score_bucket{le="60"}[5m]))
8 )
9 / sum(rate(ja4proxy_risk_score_bucket{le="100"}[5m]))

```

Type	Gauge	Table 34: ja4proxy_dial_setting — current dial value gauge
Labels	None	
Description	Current value of the dial setting (0–100). Updated on every config reload.	

ja4proxy_dial_setting

Signal Metrics

ja4proxy_signal_latency_seconds

Type	Histogram	Table 35: ja4proxy_signal_latency_seconds — per-signal latency histogram
Labels	signal: tls sni tcp asn dns beacon abuseipdb rdns	
Description	Time taken (in seconds) by each signal check, including cache hits	

Example PromQL:

```

1 # 99th percentile AbuseIPDB lookup latency
2 histogram_quantile(0.99,
3   rate(ja4proxy_signal_latency_seconds_bucket{signal="abuseipdb"}[5m])
4 )
5
6 # Mean signal latency by type
7 rate(ja4proxy_signal_latency_seconds_sum[5m])
8 / rate(ja4proxy_signal_latency_seconds_count[5m])

```

ja4proxy_signal_score

Type	Gauge	Table 36: ja4proxy_signal_score — last-seen score per signal
Labels	signal: same set as above	
Description	Most recent score contributed by each signal. Useful for debugging individual signal behaviour. Updated on every config reload.	

External Service Metrics

ja4proxy_abuseipdb_lookups_total

Example PromQL:

```

1 # AbuseIPDB cache hit rate
2 rate(ja4proxy_abuseipdb_lookups_total{result="hit"}[5m])
3 / rate(ja4proxy_abuseipdb_lookups_total[5m])
4
5 # AbuseIPDB error rate alert (should be near zero)
6 rate(ja4proxy_abuseipdb_lookups_total{result="error"}[5m]) >
   0.1

```

Type	Counter	Table 37: ja4proxy_abuseipdb_lookups_total
Labels	result: hit (cache hit) miss (cache miss → API call) error (API error)	
Description	Total AbuseIPDB cache operations	

Type	Counter	Table 38: ja4proxy_rdap_lookups_total
Labels	result: hit miss error	
Description	Total RDAP cache operations	

ja4proxy_rdap_lookups_total

Infrastructure Metrics

ja4proxy_redis_operations_total

ja4proxy_config_reloads_total

Fingerprint Metrics

ja4proxy_fingerprint_seen_total

Ban Metrics

ja4proxy_ban_ttl_seconds

Alert Rule Examples

The following Prometheus alert rules provide a starting point for production alerting:

```

1 groups:
2   - name: ja4proxy
3     rules:
4       # Redis connectivity - any failure is critical
5       - alert: JA4proxyRedisErrors
6         expr: rate(ja4proxy_redis_operations_total{result="
7         error"}[2m]) > 0.5
8         for: 1m
9         labels:
10          severity: critical
11        annotations:
12          summary: "JA4proxy Redis errors ({{ $value }}/s)"
13          description: >
14            JA4proxy is experiencing Redis errors. The fail-
15            open
16            guarantee means connections are still being allowed,
17            but bans, rate limits, and signals are not being
18            applied.
19
20          # Blocking rate spike - may indicate attack or
21          misconfiguration
22          - alert: JA4proxyHighBlockRate
23            expr: |
24              rate(ja4proxy_connections_total{action="block"}[5m])
25              / rate(ja4proxy_connections_total[5m]) > 0.3

```


Type	Counter	Table 39: ja4proxy_redis_operations_total — Redis operations by command and result
Labels	command: the Redis command (e.g. GET, ZADD, XADD); result: ok or error	
Description	Total Redis operations by command and result. Filter result="error" for failures; a sustained non-zero error rate is a problem.	
Type	Counter	Table 40: ja4proxy_config_reloads_total
Labels	result: success error	
Description	Total config reload attempts. An error result means the new config was invalid and the proxy continues with the old config.	

```

22     for: 5m
23     labels:
24         severity: warning
25     annotations:
26         summary: "JA4proxy block rate > 30%"
27         description: >
28             More than 30% of connections are being blocked.
29             Verify this is not a false positive wave.
30
31     # AbuseIPDB API errors - degraded enrichment
32     - alert: JA4proxyAbuseIPDBErrors
33       expr: rate(ja4proxy_abuseipdb_lookups_total{result="
error"}[5m]) > 0.1
34       for: 5m
35       labels:
36           severity: warning
37       annotations:
38           summary: "AbuseIPDB lookup errors"
39
40     # No connections - possible proxy failure
41     - alert: JA4proxyNoConnections
42       expr: rate(ja4proxy_connections_total[5m]) == 0
43       for: 2m
44       labels:
45           severity: critical
46       annotations:
47           summary: "JA4proxy receiving no connections"

```

Grafana Dashboard Panels

The following panels are recommended for the JA4proxy operational dashboard:

Type	Counter	Table 41: ja4proxy_fingerprint_
Labels	name: human-readable fingerprint label from <code>security.fingerprint_labels</code> , e.g. Chrome 120+, Tool/Bot (TLS)	seen_total — fingerprint observation counter
Description	Total connections seen by named fingerprint category. Useful for tracking browser vs tool traffic ratios.	

Type	Histogram	Table 42: ja4proxy_ban_ttl_sec-
Labels	None	onds — ban TTL distribution histogram
Description	Distribution of ban TTLs at the time the ban is issued. Escalating bans produce higher TTL values over t	

Panel name	Visualisation	PromQL query	Table 43: Recommended Grafana dashboard panels
Connection rate by action	Stacked time series	sum by (action) (rate(ja4proxy_connections_total[1m]))	
Risk score heatmap	Heatmap	rate(ja4proxy_risk_score_bucket[1m])	
Block rate %	Gauge (0–100%)	Block / total connections	
Bypass breakdown	Pie chart	sum by (type) (rate(ja4proxy_bypass_total[5m]))	
Signal latency P99	Bar chart	P99 histogram for each signal	
Fingerprint traffic mix	Pie chart	sum by (name) (rate(ja4proxy_fingerprint_seen_total[5m]))	
Dial setting	Stat panel	ja4proxy_dial_setting	
Redis error rate	Time series	rate(ja4proxy_redis_operations_total{result="error"}[1m])	
AbuseIPDB cache hit rate	Stat panel	hit / total rate	

Redis Schema

Redis is the shared state layer. Every JA4proxy instance reads and writes the same Redis instance. All bans, rate windows, beaconing timestamps, JA4 lists, and enrichment caches live here. The local in-process cache is a performance overlay — Redis is the source of truth.

REDIS DATA STRUCTURE SELECTION is not arbitrary. Each use case has a specific rationale — wrong data structures produce incorrect sliding windows, lost updates, or excessive memory usage. This chapter documents every key pattern, its data structure, and the rationale for that choice.

Data Structure Rationale

Never Use Redis for CIDR Matching

CIDR matching in Redis requires iterating all keys matching a pattern or maintaining a sorted set of prefixes. Both approaches are $O(n)$ for large blocklists. The in-process trie (pytricia for Python, prefix trie for Go) provides $O(\log n)$ lookup and is the only correct approach. All CIDR matching is done in-process; Redis stores the list for sharing across instances and reloading after restart.

JA4 Fingerprint Lists

ja4:blacklist

At startup, the config loader reads `security.blacklist` from `proxy.yml` and executes a Redis pipeline to `SADD` all entries to `ja4:blacklist`. On hot-reload, entries removed from config are `SREM`'d. An in-process set mirrors the Redis SET for zero-latency lookup on the hot path.

ja4:whitelist

IP Bans

ban:{ip}

Ban TTL follows an escalation schedule: initial TTL is `ban_duration`

Use case	Structure	Why this structure	Alternative con
JA4 blacklist/whitelist	SET	O(1) SISEMEMBER; small and static	Hash — unne head
IP bans	String + TTL	Per-key TTL required; atomic SETEX	SET — no per-
Sliding window rate limiting	Sorted Set + Lua (EVALSHA)	True sliding window; no boundary errors	INCR + EXPIRE dow only
Beaconing timestamps	Sorted Set (score=timestamp)	ZRANGEBYSCORE for time windows; ordered	List — no time
Session resumption ratio	Hash (total, resumed)	Atomic HINCRBY; two fields one key	Two strings — trips
Concurrent connection count	String (INCR/DECR)	Simple atomic counter	Hash — unnece
Return visitor tracking	Hash (first_seen, total, allowed, blocked)	Multi-field HINCRBY; atomic	Multiple keys —
Unique IP per CIDR	HyperLogLog (PFADD/PFCOUNT)	O(1), 12KB/key, 0.81% error	Hash of IPs — memory
Enrichment dedup	Bloom filter (BF.ADD/BF.EXISTS)	False positives acceptable; O(1)	SET — exact b memory
CIDR matching	In-process trie (NOT Redis)	No CIDR primitive in Redis; O(log n) trie	Redis SCAN —
GeoIP/ASN lookup	In-process mmap (NOT Redis)	MaxMind designed for mmap	Redis hash — optimisation
Cross-instance events	Stream (XADD/XREADGROUP)	Persistent, replayable; consumer groups	Pub/Sub — eph play
Analytics findings	String + TTL	Read by proxy; simple JSON blob	Stream — adds

Table 44: Redis data structure selection rationale

Key	ja4:blacklist
Type	SET
TTL	None (persistent)
Writer	Config loader at startup; Management UI
Reader	Pipeline layer 2 (JA4 blacklist bypass)
Example members	t13d190900_9dc949149365_97f8aa674fd9
Operations	SADD, SREM, SISEMEMBER, SMEMBERS

Table 45: ja4:blacklist key reference

(default 300s); subsequent bans double the TTL up to a maximum of 604800s (7 days). The current escalation level is stored in the JSON value's escalation field.

Rate Limiting Windows

rate:{strategy}:{key}

The sliding window is implemented as a Lua script that atomically:

1. Removes entries older than the window size (ZREMRANGEBYSCORE)
2. Adds the current timestamp (ZADD)
3. Returns the current window count (ZCOUNT)

Using Lua (EVALSHA) ensures the three operations are atomic

Key	ja4:whitelist
Type	SET
TTL	None (persistent)
Writer	Config loader at startup; Management UI
Reader	Pipeline layer 1b (JA4 whitelist bypass)
Example members	t13d1516h2_8daaf6152771_02713d6af862

Table 46: ja4:whitelist key reference

Key pattern	ban:{ip} where {ip} is the canonical IP string
Type	String
Value	JSON: {"banned_at": 1712232896, "reason": "rate_limit_by_ip", "ttl": 300}
TTL	ban_duration seconds (escalates on re-ban)
Writer	Action decider (ban action)
Reader	Pipeline layer o (early ban check); local LRU cache
Examples	ban:185.220.101.1, ban:2a06:98c1:3120::3
Operations	SET key value EX ttl, GET, DEL

Table 47: ban:{ip} key reference

from Redis's perspective — no race condition is possible between the remove and count steps.

Beaconing Data

beacon:{ip}:{ja4}

Connection State

conn:{ip}

visit:{ip}

sess:{ip}

HyperLogLog CIDR Density

hll:cidr24:{cidr} and hll:cidr48:{cidr}

HyperLogLog is the only correct data structure for this use case. Tracking all distinct IPs in a /24 in a Hash would require up to 256 entries per key; in a /48 it would be unbounded. HyperLogLog provides an accurate-enough count with fixed memory cost.

Enrichment Caches

abuseipdb:{ip}

rdap:{ip_or_prefix}

geo:{ip}

Analytics Infrastructure

events — The Event Stream

Every connection that completes the scoring pipeline produces one

Key pattern	rate:{strategy}:{key}	Table 48: rate:{strategy}:{key} key reference
Type	Sorted Set	
Score	Unix timestamp in milliseconds	
Member	Connection UUID (ensures uniqueness)	
TTL	Managed by ZREMRANGEBYSCORE; no explicit key TTL	
Writer	Pipeline rate-limiting layer	
Reader	Pipeline rate-limiting layer (ZCOUNT)	
Example keys	rate:by_ip:185.220.101.1 rate:by_ja4:t13d190900... rate:by_ip_ja4_pair:185.220.101.1:t13d190900...	
Operations	Lua EVALSHA for atomic ZADD + ZREMRANGEBYSCORE + ZCOUNT	
Key pattern	beacon:{ip}:{ja4}	Table 49: beacon:{ip}:{ja4} key reference
Type	Sorted Set	
Score	Unix timestamp in milliseconds	
Member	Unix timestamp (same as score — member must be unique)	
TTL	Managed by ZREMRANGEBYSCORE (30-minute sliding window)	
Writer	Beaconing detector signal	
Reader	Beaconing detector signal (ZRANGEBYSCORE)	
Example key	beacon:185.220.101.1:t13d1900...	

stream entry. Bypassed connections (ALPN, whitelist) also produce a minimal entry for volume tracking. The analytics node processes entries in batch using consumer groups, allowing multiple analytics workers to process in parallel without duplicate processing.

Why Streams not Pub/Sub

Redis Pub/Sub is ephemeral — a message published while a subscriber is down is lost. Redis Streams persist messages and allow consumer groups with acknowledgements. After a restart or deployment, the analytics node replays missed events from where it left off. This is critical for campaign detection, which requires historical context.

findings:{ip}

Management and Audit

management:policy_audit

Every change to `security_policy` is written here. Changes via the Management UI are attributed to the operator's session IP. Changes via `SIGHUP` config reload are attributed to "config_reload". This list provides the complete audit trail for compliance purposes.

Key Namespace Summary

Key pattern	conn:{ip}	Table 50: conn:{ip} key reference — concurrent connections
Type	String (integer)	
TTL	Managed by INCR/DECR lifecycle; expires after last connection closes	
Writer	TCP connection handler (INCR on accept, DECR on close)	
Reader	TCP behaviour signal	
Example	conn:185.220.101.1 → "42"	
Key pattern	visit:{ip}	Table 51: visit:{ip} key reference — return visitor tracking
Type	Hash	
Fields	first_seen (Unix timestamp), total (integer), allowed (integer), blocked (integer)	
TTL	None (persistent for now; rotation policy TBD)	
Writer	Action decider (HINCRBY on each action)	
Reader	TCP behaviour signal	
Key pattern	sess:{ip}	Table 52: sess:{ip} key reference — session resumption ratio
Type	Hash	
Fields	total (integer), resumed (integer)	
TTL	None (persistent)	
Writer	TLS handshake analyser	
Reader	TCP behaviour signal (ratio = resumed/total)	
Key patterns	hll:cidr24:{cidr} (IPv4 /24), hll:cidr48:{cidr} (IPv6 /48)	Table 53: HyperLogLog CIDR density keys
Type	HyperLogLog	
Operations	PFADD (add IP), PFCOUNT (count unique IPs)	
Memory	Fixed 12KB per key, regardless of distinct IP count	
Error rate	0.81% standard error	
Example key	hll:cidr24:185.220.101.0/24	
Writer	IP normalisation layer	
Reader	Analytics node (CIDR density signals)	
TTL	30 days	
Key pattern	abuseipdb:{ip}	Table 54: abuseipdb:{ip} key reference
Type	String	
Value	JSON: {"score": 85, "cached_at": 1712232896, "categories": [14, 21]}	
TTL	abuseipdb.cache_ttl seconds (default 3600)	
Writer	AbuseIPDB signal (async, after API response)	
Reader	AbuseIPDB signal (cache check)	
Key pattern	rdap:{ip_or_prefix}	Table 55: rdap:{ip_or_prefix} key reference
Type	String	
Value	JSON: {"org": "Example ISP", "prefix": "185.220.0.0/16", "country": "DE"}	
TTL	86400 seconds (24h)	
Writer	RDAP enrichment signal	
Reader	RDAP enrichment signal (cache check)	
Example key	rdap:185.220.101.1, rdap:185.220.0.0/16	
Key pattern	geo:{ip}	Table 56: geo:{ip} key reference — GeoIP country cache
Type	String	
Value	ISO 3166-1 alpha-2 country code, e.g. "RU"	
TTL	86400 seconds (24h)	
Writer	GeoIP lookup layer	
Reader	GeoIP country block layer	

Key	events	Table 57: events Redis Stream key reference
Type	Stream	
Writer	Proxy (XADD, fire-and-forget)	
Reader	Analytics node (XREADGROUP with consumer group analytics)	
Retention	MAXLEN 100000 entries (approximate, using XADD MAXLEN ~)	
Operations	XADD events * field1 val1 ... (write), XREADGROUP GROUP analytics consumer1 COUNT 100 STREAMS events >	

Key pattern	findings:{ip}	Table 58: findings:{ip} key reference — analytics findings feed-back
Type	String	
Value	JSON: {"campaign": "scan-wave-14", "attribution_score": 80, "coordinated": true}	
TTL	300 seconds	
Writer	Analytics node	
Reader	Pipeline layers 11 and 12 (attacker attribution, behavioural analysis)	

Key	management:policy_audit	Table 59: management:policy_audit key reference
Type	LIST	
Value	JSON entries: {"timestamp": "...", "changed_by": "admin@10.0.4.2", "item": "dial", "old_value": 0	
TTL	None (persistent; max 1000 entries via LTRIM)	
Writer	Management UI (config changes); config reload process	
Reader	Management UI audit log view	
Operations	LPUSH (prepend new entry), LTRIM 0 999 (keep last 1000)	

Prefix	Type	TTL	Purpose	Table 60: Complete Redis key namespace
ja4:blacklist	SET	None	JA4 fingerprint blacklist	
ja4:whitelist	SET	None	JA4 fingerprint whitelist	
ban:{ip}	String	300s–604800s	IP ban with escalating TTL	
rate:{strategy}:{key}	Sorted Set	Managed	Sliding window rate limits	
beacon:{ip}:{ja4}	Sorted Set	Managed	Beaconing IAT timestamps	
conn:{ip}	String (int)	Managed	Concurrent connection count	
visit:{ip}	Hash	None	Return visitor history	
sess:{ip}	Hash	None	Session resumption ratio	
hll:cidr24:{cidr}	HyperLogLog	30d	Unique IPs per /24 (IPv4)	
hll:cidr48:{cidr}	HyperLogLog	30d	Unique IPs per /48 (IPv6)	
abuseipdb:{ip}	String (JSON)	3600s	AbuseIPDB score cache	
rdap:{ip}	String (JSON)	86400s	RDAP org lookup cache	
geo:{ip}	String	86400s	GeoIP country cache	
events	Stream	MAXLEN 100k	Proxy event stream	
findings:{ip}	String (JSON)	300s	Analytics findings feed-back	
management:policy_audit	List	None (max 1000)	Policy change audit log	

Log Format Reference

JA4proxy emits structured JSON logs to stdout. Every connection produces exactly one connection log entry at the action decision point. Additional entries are emitted for config reloads, errors, and policy events.

JSON IS THE PRIMARY LOG FORMAT. The human-readable text format is a secondary view, derived from the JSON, suitable for interactive terminal use. All SIEM integrations, log aggregation (Loki), and alerting pipelines should use the JSON format.

JSON Log Schema

Complete Field Reference

The signals Object

When present, the `signals` field contains the individual score contribution from each signal:

```
1  "signals": {  
2    "tls":      0,  
3    "snj":      0,  
4    "tcp":      0,  
5    "asn":     15,  
6    "dns":      0,  
7    "beacon":   0,  
8    "abuseipdb": 0,  
9    "rdap":     0  
10 }
```

Each value is the raw signal score (0–100) before confidence weighting. A score of 0 means either the signal found nothing suspicious, or the external service was unavailable (fail open — both produce 0).

Example Log Entries

Allowed Browser Connection (ALPN bypass)

```
1  {  
2    "timestamp": "2026-04-04T12:34:56.789Z",  
3    "event": "connection",
```

```

4  "action": "allow",
5  "client_ip": "172.19.0.10",
6  "country": "IE",
7  "ja4": "t13d1113h2_8daaf6152771_02713d6af862",
8  "fingerprint_name": "Browser (TLS 1.3)",
9  "tls_version": "TLS 1.3",
10 "sni": "example.com",
11 "alpn": "h2",
12 "dial": 0,
13 "bypass": "alpn_browser",
14 "reason": "Browser ALPN bypass",
15 "duration_ms": 0.8
16 }

```

Note: `risk_score` and `signals` are absent because the connection was bypassed before scoring.

Blocked Connection (Spamhaus DROP match)

```

1  {
2  "timestamp": "2026-04-04T12:35:10.421Z",
3  "event": "connection",
4  "action": "block",
5  "client_ip": "185.220.101.1",
6  "country": "DE",
7  "ja4": "t13d1900h2_8daaf6152771_97f8aa674fd9",
8  "fingerprint_name": "Tool/Bot (TLS 1.3)",
9  "tls_version": "TLS 1.3",
10 "sni": null,
11 "dial": 75,
12 "bypass": "spamhaus",
13 "reason": "Spamhaus DROP match: 185.220.96.0/21",
14 "duration_ms": 0.3
15 }

```

Scored and Tarpitted Connection

```

1  {
2  "timestamp": "2026-04-04T12:36:42.100Z",
3  "event": "connection",
4  "action": "tarpit",
5  "client_ip": "45.142.212.100",
6  "country": "RU",
7  "ja4": "t13d190900_9dc949149365_97f8aa674fd9",
8  "ja4x": "a5c0d0f3d1b2_0e8a7c6b5d4e",
9  "ja4t": "8192_2_1460_nWS",
10 "fingerprint_name": "Sliver C2",
11 "tls_version": "TLS 1.3",
12 "sni": "cdn-185.example.com",
13 "risk_score": 87,
14 "dial": 75,
15 "reason": "Score 87 exceeds tarpit threshold",
16 "duration_ms": 2.4,

```

```

17 "signals": {
18   "tls": 0,
19   "sni": 40,
20   "tcp": 25,
21   "asn": 65,
22   "dns": 35,
23   "beacon": 50,
24   "abuseipdb": 80,
25   "rdap": 60
26 }
27 }

```

Banned Connection (Subsequent Request)

```

1 {
2   "timestamp": "2026-04-04T12:37:05.002Z",
3   "event": "connection",
4   "action": "block",
5   "client_ip": "45.142.212.100",
6   "country": "RU",
7   "dial": 75,
8   "bypass": "ban_cache",
9   "reason": "Banned for 604800s (escalation level 3)",
10  "duration_ms": 0.1
11 }

```

Subsequent connections from a banned IP are rejected at layer 0 from the local ban cache. No JA4 parsing occurs, hence the absent fingerprint fields.

Config Reload Event

```

1 {
2   "timestamp": "2026-04-04T13:00:00.000Z",
3   "event": "config_reload",
4   "action": null,
5   "result": "success",
6   "changed_keys": ["dial", "security_policy.spamhaus_bypass.enabled"],
7   "duration_ms": 45.2
8 }

```

Bypass Disabled Warning

```

1 {
2   "timestamp": "2026-04-04T08:00:00.001Z",
3   "event": "bypass_disabled",
4   "bypass": "alpn_browser_bypass",
5   "effect": "browser traffic will be scored; false positive risk elevated",
6   "level": "WARN"

```

```
7 }
```

Human-Readable Text Format

The text format is a condensed single-line summary derived from the JSON event. It is written to a separate log stream (or can be toggled on/off) for operator convenience during interactive sessions.

```
1 ALLOWED: 172.19.0.10 | Country: IE | JA4: t13d1113h2_... |
   Name: Browser (TLS 1.3) | TLS: TLS 1.3
2 BLOCKED: 185.220.101.1 | Country: DE | Reason: Spamhaus DROP:
   185.220.96.0/21
3 TARPIT: 45.142.212.100 | Country: RU | JA4: t13d190900_... |
   Name: Sliver C2 | Score: 87
4 BANNED: 45.142.212.100 | Country: RU | Reason: Banned for
   604800s (level 3)
```

Log Levels

Loki Log Queries (LogQL)

Assuming logs are shipped to Loki with the label `job="ja4proxy"`, the following LogQL queries cover common operational needs:

```
1 # All blocked connections in last hour
2 {job="ja4proxy"} | json | action="block"
3
4 # All connections from a specific IP
5 {job="ja4proxy"} | json | client_ip="185.220.101.1"
6
7 # Connections with risk_score > 80
8 {job="ja4proxy"} | json | risk_score > 80
9
10 # All Spamhaus bypasses
11 {job="ja4proxy"} | json | bypass="spamhaus"
12
13 # Tarpitted connections with their signal breakdown
14 {job="ja4proxy"} | json | action="tarpit"
15 | line_format "{{.client_ip}} score={{.risk_score}}"
16
17 # Config reload events
18 {job="ja4proxy"} | json | event="config_reload"
19
20 # Error events only
21 {job="ja4proxy"} | json | level="ERROR"
22
23 # Fingerprint distribution for last 24h
24 {job="ja4proxy"} | json | fingerprint_name != ""
25 | unwrap risk_score
26 | rate by (fingerprint_name) [5m]
```

SIEM Integration — CEF Format

For SIEM platforms that require Common Event Format (CEF), the following mapping converts JA4proxy JSON fields to CEF:

Example CEF line for a tarpitted connection:

```
1 CEF:0|JA4proxy|JA4proxy|1.0|connection_tarpit|connection_tarpit
   |8|
2   src=45.142.212.100 act=tarpit cs1Label=JA4
3   cs1=t13d190900_9dc949149365_97f8aa674fd9
4   cs2Label=Country cs2=RU cs3Label=Reason
5   cs3=Score 87 exceeds tarpit threshold cn1Label=Score
6   cn1=87 cn2Label=Dial cn2=75
7   start=2026-04-04T12:36:42.100Z
```

Field	Type	Always present	Description	Table 61: JSON log field reference — connection event
timestamp	string (ISO 8601)	Yes	UTC timestamp with milliseconds: 2026-04-04T12:34:56.789Z	
event	string	Yes	Event type: connection, config_reload, error, bypass_disabled	
action	string	Yes	Final action: allow, block, tarpit, ban, flag, rate_limit	
client_ip	string	Yes	Canonical client IP (from PROXY protocol). Full IPv4 or IPv6 string.	
country	string	No	ISO 3166-1 alpha-2 country code. Absent if GeoIP lookup failed.	
ja4	string	No	JA4 fingerprint string. Absent if ClientHello could not be parsed.	
ja4x	string	No	JA4X extended fingerprint. Absent if not computed.	
ja4t	string	No	JA4T TCP fingerprint. Absent if not computed.	
fingerprint_name	string	No	Human-readable label for JA4, from <code>security.fingerprint_labels</code>	
tls_version	string	No	TLS version string: TLS 1.3, TLS 1.2, etc.	
sni	string	No	Server Name Indication from ClientHello. Absent if not present in handshake.	
alpn	string	No	First ALPN value: h2, h1, etc.	
risk_score	integer	No	Computed risk score (0–100). Absent for bypassed connections.	
dial	integer	Yes	Current dial setting at decision time.	
bypass	string	No	Bypass type if connection was bypassed: alpn_browser, ja4_whitelist, etc.	
reason	string	Yes	Human-readable reason for the action.	
duration_ms	float	Yes	Pipeline execution time in milliseconds.	
signals	object	No	Per-signal scores as a JSON object. Absent for bypassed connections.	

Level	Volume	Emitted for	Table 62: Log levels and when each is emitted
ERROR	Low	Redis connection failure, config parse failure, TLS parser crash, external API repeated failure	
WARN	Low	Bypass disabled at startup, AbuseIPDB rate limit hit, RDAP block expansion triggered	
INFO	Medium	Config reload (success or error), startup completion, shutdown	
DEBUG	High	Individual signal scores, cache hits/misses, Redis operations. Disabled in production.	
(connection)	Very high	Every connection event — not strictly a log level; always emitted	

CEF field	JA4proxy JSON field	Notes	Table 63: CEF field mapping for JA4proxy connection events
CEF:0 JA4proxy JA4proxy 1.0	—	CEF header (vendor product version)	
name	event + action	e.g. connection_block	
severity	risk_score / 10	Scale: 0–10	
src	client_ip	Source IP	
cs1 (JA4)	ja4	Custom string field	
cs2 (Country)	country	Custom string field	
cs3 (Reason)	reason	Custom string field	
cn1 (Score)	risk_score	Custom number field	
cn2 (Dial)	dial	Custom number field	
act	action	Device action	
start	timestamp	Event time	
deviceProcessName	"ja4proxy"	Fixed string	

Deployment Reference

JA4proxy is deployed as a set of Docker containers orchestrated by Docker Compose (development and single-host production) or Helm (Kubernetes multi-host production). This chapter documents every service, environment variable, port, and network requirement.

THE REFERENCE DEPLOYMENT TOPOLOGY places JA4proxy between a HAProxy load balancer and the backend origin server. All traffic from the internet passes through HAProxy (:443), which forwards raw TCP streams (with PROXY protocol v2 headers) to JA4proxy instances (:8080). JA4proxy then proxies allowed connections to the backend (:443).

Docker Compose Reference

Services Overview

Service	Image	Ports	Role
haproxy	haproxy:2.9-alpine	0.0.0.0:443:443	TLS passthrough LB; injects PROXY protocol v2
ja4proxy	ghcr.io/seanpor/ja4proxy-go:main	0.0.0.0:8443:8443	JA4proxy pipeline; exposes metrics on :9090
management	ghcr.io/seanpor/ja4proxy-management:main	127.0.0.1:8090:8090	Management API and dashboard (plain HTTP, loopback only)
management-tls	caddy:2-alpine	127.0.0.1:8444:8444	HTTPS sidecar for management dashboard; auto-generates self-signed cert
redis	redis:7-alpine	127.0.0.1:6379:6379	Shared state; NOT exposed to internet
analytics	ja4analytics:latest	—	Redis Streams consumer; findings writer
prometheus	prom/prometheus:v2.51.0	127.0.0.1:9090:9090	Metrics collection
grafana	grafana/grafana:10.4.0	127.0.0.1:3000:3000	Dashboards
loki	grafana/loki:2.9.5	127.0.0.1:3100:3100	Log aggregation
alertmanager	prom/alertmanager:v0.27.0	127.0.0.1:9093:9093	Alert routing
backend	(operator-provided)	—	Protected origin; only reachable from proxy

Port Binding Policy

Management services (Prometheus, Grafana, Loki, Alertmanager) bind to 127.0.0.1 only. They must never be exposed on 0.0.0.0 in production. Redis binds to 127.0.0.1. Only HAProxy binds to 0.0.0.0:443. The proxy service binds to 127.0.0.1:8080 — it receives connections only from HAProxy, which is on the same host or the same Docker network.

Docker Networks

Network	Subnet	Connected services	Purpose
dmz	172.20.0.0/24	haproxy, proxy	HAProxy → proxy path
app	172.21.0.0/24	proxy, redis, analytics, backend	proxy → redis, backend
monitoring	172.22.0.0/24	proxy, prometheus, loki	Metrics and log scraping
mgmt	172.23.0.0/24	prometheus, grafana, alertmanager, loki	Observability stack internal

Table 65: Docker Compose network definitions

Environment Variables

The following environment variables must be set in the deployment environment (e.g. in a `.env` file alongside `docker-compose.yml`).

Variable	Required	Default	Description
REDIS_PASSWORD	Yes	—	Redis AUTH password. Set on both proxy and redis services.
ABUSEIPDB_API_KEY	No	—	AbuseIPDB API key. If unset, AbuseIPDB lookups are disabled.
JA4_DIAL	No	0	Override dial setting at container start (useful for testing).
JA4_LOG_LEVEL	No	INFO	Log verbosity: DEBUG, INFO, WARN, ERROR
JA4_BACKEND_HOST	No	backend	Backend hostname override. Useful when backend DNS differs.
JA4_BACKEND_PORT	No	443	Backend port override.
GF_SECURITY_ADMIN_PASSWORD	No	admin	Grafana admin password. Change before production deployment.
GF_SERVER_ROOT_URL	No	—	Grafana public URL for link generation.
MAXMIND_LICENSE_KEY	Conditional	—	MaxMind license key for GeoLite2 database download. Required if using auto-download.

Table 66: Required and optional environment variables

Service	Host path	Container path	Notes
proxy	./config	/app/config	Read-only (RO). Config files.
proxy	./certs	/app/certs	Read-only. mTLS CA and certs.
proxy	./data/geoip	/app/data/geoip	Read-only. MaxMind DB files.
redis	./data/redis	/data	Redis persistence (RDB/AOF).
management-tls	caddy_data (named volume)	/data	Caddy certificate storage.
management-tls	caddy_config (named volume)	/config	Caddy runtime config.
prometheus	./data/prometheus	/prometheus	Prometheus TSDB.
grafana	./data/grafana	/var/lib/grafana	Grafana dashboards and state.
loki	./data/loki	/loki	Loki log chunks.

Table 67: Docker volume mounts

Volume Mounts

HAProxy Configuration

HAProxy must be configured for TLS passthrough with PROXY protocol v2 injection. The critical sections of `haproxy.cfg`:

```

1 frontend tls_in
2     bind *:443
3     mode tcp
4     option tcplog
5     default_backend ja4proxy
6
7 backend ja4proxy
8     mode tcp
9     option tcp-check
10    server proxy1 proxy:8080 check send-proxy-v2
11    # send-proxy-v2 enables PROXY protocol v2 header injection
12    # TLS is NOT terminated - raw TCP passthrough

```

TLS Passthrough — Not TLS Termination

HAProxy must be configured in TCP mode (`mode tcp`), not HTTP mode (`mode http`). HTTP mode would terminate the TLS connection and re-encrypt it, defeating the purpose of JA4proxy. In TCP mode, HAProxy passes the raw TLS bytes through to JA4proxy unchanged. Only the PROXY protocol header is added by HAProxy.

Pre-built Container Images

The Go proxy and management API are published to the GitHub Container Registry (GHCR) on every merge to `main`. Running `docker compose pull` fetches them automatically — no build toolchain is required on the production host:

```

1 docker compose pull && docker compose up -d

```

Published image names:

- `ghcr.io/seanpor/ja4proxy-go:main`
- `ghcr.io/seanpor/ja4proxy-management:main`

Both packages are public. Images are also tagged `sha-<commit>` for pinning and `v*` on version releases.

Traffic Insertion via iptables

When JA4proxy is deployed on the same host as an existing service (nginx, HAProxy, or the application directly), it can be inserted into the live traffic path without changing any upstream configuration by using an iptables PREROUTING redirect:

```
1 # Insert JA4proxy into the traffic path (:443 -> :8443):
2 make traffic-on
3
4 # Remove JA4proxy from the traffic path (instant rollback):
5 make traffic-off
```

These targets expand to:

```
1 # traffic-on:
2 sudo iptables -t nat -I PREROUTING -p tcp --dport 443 \
3     -j REDIRECT --to-port 8443
4
5 # traffic-off:
6 sudo iptables -t nat -D PREROUTING -p tcp --dport 443 \
7     -j REDIRECT --to-port 8443
```

The redirect applies to incoming packets only (PREROUTING). Loopback traffic is unaffected. Rollback is instantaneous and requires no service restart. Requires `sudo iptables` on the host.

Helm Chart Reference

Chart Structure

The JA4proxy Helm chart is located at `helm/ja4proxy/`. It deploys:

- A Deployment for the proxy pods
- A Service (ClusterIP) exposing port 8080
- A ConfigMap containing `proxy.yml`
- A Secret for Redis password and API keys
- A ServiceMonitor for Prometheus Operator scraping
- (Optional) A HorizontalPodAutoscaler

values.yaml Reference

Kubernetes Resources Created

```
1 # Install
```

Table 68: Key Helm chart values

Key	Type	Default	Description
replicaCount	integer	2	Number of proxy pod replicas
image.repository	string	ja4proxy	Container image repository
image.tag	string	latest	Image tag. Pin to a specific version in production.
proxy.backendHost	string	<i>required</i>	Backend service hostname
proxy.backendPort	integer	443	Backend port
proxy.dial	integer	0	Initial dial setting
redis.url	string	<i>required</i>	Redis connection URL
redis.existingSecret	string	""	Kubernetes Secret name containing <code>redis-password</code> key
abuseipdb.enabled	boolean	true	Enable AbuseIPDB lookups
abuseipdb.existingSecret	string	""	Secret containing <code>api-key</code>
resources.requests.cpu	string	100m	CPU request per pod
resources.requests.memory	string	128Mi	Memory request per pod
resources.limits.cpu	string	1000m	CPU limit
resources.limits.memory	string	256Mi	Memory limit
autoscaling.enabled	boolean	false	Enable HPA
autoscaling.minReplicas	integer	2	Minimum replicas
autoscaling.maxReplicas	integer	10	Maximum replicas
autoscaling.targetCPUUtilizationPercentage	integer	70	HPA CPU target
serviceMonitor.enabled	boolean	true	Create ServiceMonitor for Prometheus Operator
geoip.volumeClaimName	string	""	PVC name for GeoIP database files (read-only mount)

```

2 helm install ja4proxy helm/ja4proxy \
3   --set proxy.backendHost=backend-svc.app.svc.cluster.local \
4   --set redis.url=redis://redis-svc.data.svc.cluster.local:6379
5   \
6   --set redis.existingSecret=ja4proxy-secrets
7 # Upgrade (e.g., raise dial)
8 helm upgrade ja4proxy helm/ja4proxy --set proxy.dial=50

```

Networking Requirements

Firewall Rules for DMZ Deployment

Production Deployment Checklist

The following checklist must be completed before any production deployment:

- ☐ **Set dial:** 0 — deploy in monitor-only mode
- ☐ **Pin image tags** — replace `latest` with a specific version
- ☐ **Change default passwords** — Grafana admin, Redis password

Source	Destination	Port	Protocol	Purpose
Table 69. Required firewall rules for DMZ deployment				
Internet	HAProxy (DMZ)	443	TCP	Incoming TLS connections
HAProxy (DMZ)	JA4proxy (App)	8080	TCP	PROXY protocol + TLS passthrough
JA4proxy (App)	Redis (Data)	6379	TCP	State read/write
JA4proxy (App)	Backend (App)	443	TCP	Forwarded TLS connections
JA4proxy (App)	DNS resolver	53	UDP/TCP	FCrDNS lookups
JA4proxy (App)	AbuseIPDB API	443	TCP	Reputation lookup (outbound)
JA4proxy (App)	RDAP endpoints	443	TCP	RDAP lookup (outbound)
JA4proxy (App)	Spamhaus URLs	443	TCP	Blocklist refresh (outbound)
Prometheus (Mgmt)	JA4proxy (App)	9090	TCP	Metrics scraping
Loki (Mgmt)	JA4proxy (App)	(log push)	TCP	Log collection
Operator VPN	Grafana (Mgmt)	3000	TCP	Dashboard access
Operator VPN	Alertmanager (Mgmt)	9093	TCP	Alert management
Blocked (explicit DENY):				
Internet	JA4proxy (App)	any	any	Proxy never directly reachable from internet
Internet	Redis (Data)	any	any	Redis never internet-accessible
Internet	Mgmt zone	any	any	Management never internet-accessible

- ☐ **Secrets via Secret Manager** — no passwords in .env committed to Git
- ☐ **TLS certificate for backend** — verify backend cert is valid and trusted
- ☐ **Trusted upstream CIDR** — configure HAProxy address range in proxy config
- ☐ **GeoIP database present** — MaxMind GeoLite2 files mounted and current
- ☐ **Redis persistence enabled** — RDB or AOF configured; data volume mounted
- ☐ **Loki log shipping** — proxy logs flowing to Loki; retention policy set
- ☐ **Prometheus scraping** — metrics endpoint reachable; dashboards loaded
- ☐ **Alert rules loaded** — minimum alert set from Chapter
- ☐ **HAProxy in TCP mode** — not HTTP mode; TLS passthrough verified
- ☐ **Management ports bound to 127.0.0.1** — not 0.0.0.0
- ☐ **Verify score distribution** — run for 24h at dial=0; review histogram
- ☐ **Phase 14 hardening** — complete Phase 14 checklist before raising dial above 0

Phase 14 Hardening Prerequisite

The Phase 14 production hardening checklist (docs/DMZ_DEPLOYMENT_READINESS.md) must be completed before raising the dial above 0 in production. Key gaps identified at POC stage include: secrets management integration, Redis TLS, rate-limit

self-protection for the tarpit mechanism, and resource limit tuning. Do not skip this step.

Security Reference

JA4proxy is a security tool, but it is also a target. This chapter documents the security properties of JA4proxy itself: what it guarantees, where its boundaries are, what its known gaps are at POC stage, and how to harden it for production.

THE SECURITY MODEL OF JA4PROXY rests on three guarantees: it never decrypts traffic (cryptographic transparency), it fails open when uncertain (false positive minimisation), and it enforces a configurable hard floor below which browser traffic can never be blocked (ALPN structural bypass).

The Bypass System — Security Model

Bypass Architecture

Bypass conditions are not simply configuration flags — they represent a deliberate security policy decision about which classes of traffic should be exempt from scoring. Each bypass type has a specific threat model and risk profile when enabled or disabled.

Startup Warnings for High-Risk Bypass Disabling

When any bypass is disabled that has elevated false positive or false negative risk, JA4proxy emits a WARN level log at startup. This ensures the operator cannot accidentally disable a bypass in a config file and have the change go unnoticed.

```
1 # Example startup warnings for disabled bypasses
2 WARN | policy | event=bypass_disabled | bypass=
   alpn_browser_bypass
3   | effect=browser traffic will be scored; false positive
   risk elevated
4
5 WARN | policy | event=bypass_disabled | bypass=
   spamhaus_drop_bypass
6   | effect=Spamhaus DROP matches scored as signal(+80)
   instead of hard block
```

Bypass	Threat addressed	Risk if disabled	Use case for disabling
alpn_browser_bypass	Not applicable — this is a false-positive protection, not a threat bypass	Browser traffic scored, significant false positive risk at scale	Scoring specific h2 API clients known to not be real browsers
ja4_whitelist_bypass	Trusted fingerprints (known-good tools, monitoring) bypass scorer	Whitelisted fingerprints may be blocked if they score above threshold	Testing whether a whitelisted fingerprint would be scored
mtls_bypass	Cryptographic identity proofing — mTLS client is verified	mTLS clients scored and potentially blocked	Auditing mTLS client behaviour
static_ip_allowlist	Trusted internal IPs (monitoring, CI) bypass scorer	Monitoring may be blocked; CI tests fail	Temporary removal during security audit
ja4_blacklist_bypass	Known-bad fingerprints (C2, scanners) immediately rejected	Blacklisted fingerprints only blocked if score exceeds dial threshold	Investigating traffic from a specific fingerprint
country_blacklist_bypass	Geographically restricted access	Blocked countries go through scorer; may not be blocked at low dial	Investigating specific country traffic
spamhaus_bypass	Hijacked/criminal IP blocks immediately rejected	Spamhaus matches produce signal(+80) instead of hard block	Investigating Spamhaus-listed range
tls_version_bypass	Deprecated TLS versions immediately rejected	Old TLS produces risk signal instead of hard reject	Auditing legacy client population

The Asymmetry Principle — In Depth

Why Fail Open is Correct

The asymmetry principle (Chapter) has specific technical manifestations throughout the system. Understanding why each mechanism exists as it does requires understanding the cost model:

- **A false negative** (bad bot allowed) produces a malformed request to the backend. The backend logs it. The analytics node may detect the pattern retrospectively. The next connection from the same IP will have a better signal set (AbuseIPDB cache populated, beaconing window extended). Recovery is automatic.
- **A false positive** (real browser blocked) produces a failed page load for a human user. The user sees a connection error with no explanation. They may never return. There is no automatic recovery — the damage is immediate and irreversible.

False Positive Protection Mechanisms

These mechanisms exist specifically to prevent false positives:

ALPN structural bypass h2/h1 ALPN connections can never be blocked, regardless of score or dial setting. This is not a configurable threshold — it is structural. Even with dial=100 and all signals firing, a connection presenting h2 ALPN is allowed. Only if the operator explicitly disables this bypass does the structural protection lift.⁵

⁵ Browser traffic presenting h2 or h1 ALPN could theoretically score highly if, for example, the IP is on Spamhaus DROP and the browser is behind a compromised ISP. The bypass ensures this never causes a false positive.

ALLOW cached 30min, BLOCK cached 30s The local LRU cache intentionally uses asymmetric TTLs. An IP allowed 29 minutes ago gets another 30-minute free pass. An IP blocked 31 seconds ago is re-evaluated against current Redis state.

Local cache wins on Redis failure When the local cache says allow and Redis says block (or Redis is unreachable), the local cache wins. Real users in the local cache continue to be served even during Redis outages.

External service unavailability → zero signal If AbuseIPDB is down, RDAP is unreachable, or DNS lookups time out, the affected signal contributes 0 to the score. The score can only go down when services fail, not up.

AbuseIPDB floor at 50 AbuseIPDB scores below 50 are suppressed. This prevents shared infrastructure (NAT gateways, corporate egress, VPNs) from triggering signals just because one user behind that NAT misbehaved.

RDAP block expansion off by default Expanding a block to an entire /24 based on one bad IP is a high-impact action. The default is off. Operators must consciously opt in.

The Fail-Open Guarantee

The fail-open guarantee is: **in all uncertainty, connections are allowed rather than blocked**. This guarantee applies to:

- Redis unreachable: bans not enforced; rate limits not enforced; signals that depend on Redis return 0
- External service (AbuseIPDB, RDAP, DNS) unreachable: affected signals return 0
- Config parse error on reload: previous valid config retained; new config not applied
- Pipeline exception: connection is allowed; exception is logged
- Unknown fingerprint: score 0 (unknown is not inherently suspicious)

Fail-Open is Not a Bug

Operators sometimes ask whether fail-open can be changed to fail-closed. The answer is no — this would be an architectural change that fundamentally compromises the false positive guarantee. Fail-closed means an outage blocks real users. The system is designed to tolerate outages gracefully, not to use them as an excuse to block.

Hot Config Reload — Security Implications

Hot config reload is a convenience, but it has security implications:

Policy change audit log Every config reload is written to `management:policy_audit` in Redis with a timestamp and attribution. Operators can audit all policy changes without consulting application logs.

Config file integrity The proxy does not verify the cryptographic integrity of `config/proxy.yml` before loading. In production, config files should be managed by a configuration management system (Ansible, Terraform, Kubernetes ConfigMap) with change control and audit trail.

SIGHUP race condition There is a brief window during config reload where new bypass states are being applied and some connections may be evaluated under the old configuration. This window is bounded by the duration of the reload operation (typically <50ms). No connections are dropped during reload.

Management UI reload The Management UI triggers reload via a Redis pub/sub message. If Redis is compromised, an attacker could trigger arbitrary config reloads. Redis authentication (`redis.password`) must be set in production.

Phase 200-Series Hardening

THE PHASE 200-SERIES CLOSES THE PRODUCTION-HARDENING GAPS flagged in Phase 14 and the comprehensive security audit. These items are now in production builds of the Go proxy.

Redis TLS (Phase 201)

Redis communications are now TLS-encrypted by default. The Go proxy connects to Redis with mutual TLS where the Redis instance is configured with `tls-port` and a server certificate; client certificates are loaded from the path configured under `redis.tls.client_cert_path`. Plaintext Redis connections remain available for development (a Unix-socket-only configuration with restrictive filesystem permissions is the recommended local-dev path), but production Helm charts default

to TLS-on. See `docs/phases/PHASE_201.md` for the full configuration matrix.

PROXY Protocol v2 with TLVs (Phase 203)

The Go proxy now parses PROXY protocol v2 binary headers, including TLV (type-length-value) extensions. The TLV trust path is gated by `security_policy.trusted_upstream_cidrs`: a TLV is honoured only when the TCP peer is within the configured trusted CIDR. This prevents a direct-to-port-8080 attacker from forging TLVs to claim a whitelisted source IP. The trust check runs before any TLV is parsed, so a malformed TLV from an untrusted peer cannot trigger a parser path at all.

Default Credentials Removed (Phase 202)

Default credentials have been removed from the Docker image, Helm chart, and `docker-compose.yml`. Previously, Redis shipped with `requirepass`, Grafana shipped with `admin/admin`, and the Management API had a hard-coded local-dev token. All three now require explicit env-var provisioning at startup; the proxy refuses to start if a required credential is unset. The Helm chart fails its preflight check and Docker Compose surfaces a clear error rather than silently starting with a blank password. See `docs/phases/PHASE_202.md` for the env-var reference and the migration runbook.

Known Security Gaps (POC Stage)

The following gaps are documented in `docs/DMZ_DEPLOYMENT_READINESS.md` and `docs/security/COMPREHENSIVE_SECURITY_AUDIT.md`. They must be addressed before production deployment with dial > 0.

Production Hardening Checklist

- ☐ **Secrets manager integration** — API keys and passwords from Vault or cloud secret manager; never in `.env` files
- ☐ **Redis TLS** — enable TLS for Redis connections or use Unix socket with restricted permissions
- ☐ **Redis AUTH** — set strong Redis password; verify it is not default
- ☐ **Redis ACLs** — separate credentials for proxy, analytics, and management services
- ☐ **Management UI authentication** — complete Phase 13 before deploying Management UI
- ☐ **Tarpit resource limits** — configure maximum concurrent tarpitted connections
- ☐ **Container non-root user** — verify proxy runs as non-root (uid 1000)
- ☐ **Read-only filesystem** — mount application directories read-only where possible

Gap	Severity	Phase addressing it	Description
Secrets in environment variables	Medium	Phase 14	API keys and Redis password should come from a secret manager (Vault, AWS Secrets Manager), not plain environment variables
Redis without TLS	High	Phase 14	Redis communications are unencrypted. In production, Redis must use TLS or a Unix domain socket with filesystem permissions
Tarpit self-protection absent	Medium	Phase 14	The tarpit mechanism has no limit on concurrent tarpitted connections. A sustained flood could exhaust file descriptors
Config file not integrity-checked	Low	Phase 14	No signature verification on proxy.yml before hot-reload
Management UI authentication	High	Phase 13	Management UI has no authentication in POC. Must add before production
Redis ACLs not configured	Medium	Phase 34	All clients share a single Redis connection. Should use ACLs with per-service credentials
No Seccomp/AppArmor profiles	Low	Phase 55	Container syscall surface not restricted
Supply chain integrity	Low	Phase 35	No SBOM, no container image signing

- ☐ **Network policy** — enforce network zone isolation (see Chapter)
- ☐ **HAProxy trusted CIDR** — configure the exact HAProxy IP range in proxy trusted upstream config
- ☐ **Grafana admin password changed** — default admin/admin must not be used
- ☐ **Loki log retention** — configure retention policy (30 days recommended for GDPR compliance)
- ☐ **Audit log retention** — verify management:policy_audit retention policy
- ☐ **Pentest validation** — complete the adversarial test suite (Phase 45) before raising dial

IP Spoofing Protection

Without trusted upstream CIDR verification, an attacker who can reach port 8080 directly could inject a PROXY protocol header claiming any source IP address — effectively bypassing all IP-based checks by claiming to be a whitelisted IP.

JA4proxy mitigates this by verifying that the sender of the TCP

connection (before reading the PROXY protocol header) is within the configured trusted upstream CIDR. Only HAProxy addresses should be in this CIDR. If the sender is not in the trusted CIDR, the PROXY protocol header is rejected and the real TCP peer IP is used instead.

Port 8080 Must Not Be Internet-Accessible

Port 8080 (the JA4proxy listen port) must only be reachable from HAProxy. If an attacker can reach port 8080 directly, they can manipulate their apparent source IP. Firewall rules must block port 8080 from all sources except the HAProxy IP range.

JA4 Fingerprint Limitations

JA4 fingerprinting is powerful but not infallible. Security engineers should understand its limitations:

JA4 copying An attacker can copy a known-good JA4 fingerprint by configuring their TLS library to match the exact cipher suite list and extension ordering. JA4T (TCP fingerprint) makes this harder — it requires matching the OS-level TCP stack, not just the TLS library.

JA4 rotation A sophisticated attacker can rotate through JA4 values to evade per-fingerprint rate limits. The `by_ip_ja4_pair` rate limit strategy is designed to catch this pattern.

JA4 collisions Multiple legitimate clients may share the same JA4 value. JA4 blacklisting should only be applied to fingerprints that are *uniquely* associated with malicious tools — not fingerprints shared with any legitimate client software.

Compliance Reference

JA4proxy processes IP addresses and TLS fingerprints in the course of its security function. This chapter documents what personal data is processed, for how long, what controls apply, and how to satisfy data subject requests.

COMPLIANCE OBLIGATION DEPENDS ON DEPLOYMENT CONTEXT. JA4proxy is a security proxy; the operator who deploys it is the data controller. This chapter provides the technical facts that a data controller needs to assess their compliance obligations. It does not constitute legal advice.

Data Inventory

What Data is Processed

Data element	Personal data?	Table 72: Data elements processed by JA4proxy	Where stored
Client IP address (IPv4/IPv6)	Yes (see note)		Redis, logs, Prometheus labels
JA4 fingerprint	No (device, not person)		Redis, logs
JA4X fingerprint	No		Redis, logs
JA4T fingerprint	No		Redis, logs
SNI hostname	No (operator’s domain)		Logs
Country code	No (GeoIP; not precise location)		Redis, logs
ASN organisation name	No		Logs (transient)
AbuseIPDB score	Derivative of IP (see note)		Redis (cache)
RDAP organisation	No		Redis (cache)
PTR/FCrDNS hostname	Potentially (if ISP assigns personal hostnames)		In-process cache (transient)
Beaconing timestamps	Linked to IP (see note)		Redis Sorted Set
Connection metadata (port, timing)	Potentially linked to IP		Logs, Prometheus

IP Addresses as Personal Data

Under GDPR Recital 30, IP addresses are personal data when they can be linked to a natural person. Dynamic IP addresses assigned by ISPs to residential customers are generally considered personal data in the EU. Static IP addresses assigned to organisations are generally not personal data. JA4proxy pro-

cesses both categories and cannot distinguish them at the time of processing. Operators must account for this in their DPIA.

Legal Basis for Processing

The operator must establish a legal basis for processing IP addresses. Typical bases:

Legitimate interests (Art. 6(1)(f)) Security monitoring and DDoS/bot mitigation is a legitimate interest of the controller. The processing is proportionate because: only metadata is processed (no content decryption), data is retained for the minimum period necessary, and the security purpose is genuine.

Legal obligation (Art. 6(1)(c)) If the operator is subject to regulations requiring security controls (PCI-DSS, NIS2), processing for compliance purposes has a legal basis.

Data Retention

Redis Key TTLs

Key type	Contains IP?	TTL	Notes
ban:{ip}	Yes	300s–604800s	Escalating ban. Max 7 days.
rate:{strategy}:{key}	Yes (in key)	60s window	Sliding window; entries expire with window
beacon:{ip}:{ja4}	Yes (in key)	30min window	Timestamps expire; key persists until window
conn:{ip}	Yes (in key)	Connection lifetime	Deleted when connection count reaches limit
visit:{ip}	Yes (in key)	None (persistent)	Manual deletion required for erasure
sess:{ip}	Yes (in key)	None (persistent)	Manual deletion required for erasure
hll:cidr24:{cidr}	Yes (CIDR containing IP)	30 days	Approximate; cannot extract individual IPs
abuseipdb:{ip}	Yes (in key)	3600s	Cache; auto-expires
geo:{ip}	Yes (in key)	86400s	Cache; auto-expires
rdap:{ip}	Yes (in key)	86400s	Cache; auto-expires
findings:{ip}	Yes (in key)	300s	Analytics output; auto-expires
management:policy_audit	No	None (max 1000 entries)	Audit log; no IP data

Log Retention

Connection logs are emitted to stdout and shipped to Loki (or another log aggregator). The retention period is configured in the log aggregation system:

- **Recommended retention:** 30 days for operational logs
- **Security incident retention:** 90 days (check applicable law)
- **Legal hold:** Operator-specific; coordinate with legal team

Log retention is configured in Loki's `storage_config` section. JA4proxy does not enforce log retention directly — this is the responsibility of the log aggregation pipeline.

Prometheus Metrics Retention

Prometheus metrics contain IP addresses only in specific label combinations (notably the ban metrics, which label by IP). Default Prometheus retention is 15 days. For compliance, either:

1. Reduce retention to 7 days for metrics containing IP labels
2. Configure remote write to a long-term storage system with access controls
3. Consider using IP prefix labels (/24 or /48) instead of full IPs for metrics

Right to Erasure (GDPR Art. 17)

Data Subject Access Request (DSAR) Process

When a data subject submits a right-to-erasure request identifying their IP address, the operator must delete all personal data associated with that IP. The following Redis keys must be deleted:

```

1  # DSAR erasure script for a specific IP address
2  # Replace TARGET_IP with the subject's IP address
3
4  TARGET_IP="203.0.113.42"
5
6  # Delete all Redis keys containing this IP
7  redis-cli DEL "ban:${TARGET_IP}"
8  redis-cli DEL "conn:${TARGET_IP}"
9  redis-cli DEL "visit:${TARGET_IP}"
10 redis-cli DEL "sess:${TARGET_IP}"
11 redis-cli DEL "abuseipdb:${TARGET_IP}"
12 redis-cli DEL "geo:${TARGET_IP}"
13 redis-cli DEL "rdap:${TARGET_IP}"
14 redis-cli DEL "findings:${TARGET_IP}"
15
16 # Delete beaconing keys (need to enumerate JA4 variants)
17 redis-cli SCAN 0 MATCH "beacon:${TARGET_IP}:*" COUNT 100
18 # Delete each matched key
19
20 # Delete rate limit keys
21 redis-cli SCAN 0 MATCH "rate:*:${TARGET_IP}*" COUNT 100
22 # Delete each matched key
23
24 # Logs: use log aggregation platform API to delete/redact log
    entries
25 # HyperLogLog: individual IPs cannot be removed from HLL -
    document this limitation

```

HyperLogLog Erasure Limitation

HyperLogLog data structures (hll:cidr24: and hll:cidr48:) do not support removal of individual elements. Once an IP has been added to a HLL, it cannot be removed without deleting

the entire HLL key (which would also remove all other IPs from that CIDR). Document this technical limitation in your DPIA and assess whether HLL data constitutes personal data under your jurisdiction (it does not reveal the actual IP address, only an approximate count).

IP Anonymisation Option

For environments where storing full IP addresses is not appropriate, JA4proxy can be configured to truncate IP addresses in logs and Redis keys:

- IPv4: truncate last octet (e.g. 203.0.113.42 → 203.0.113.0)
- IPv6: truncate to /48 (e.g. 2001:db8::1 → 2001:db8::)

IP Anonymisation Degrades Security Function

Truncating IP addresses prevents ban enforcement (you cannot ban 203.0.113.0/24 without blocking 255 potentially innocent IPs). It also prevents return visitor tracking, session resumption ratio, and concurrent connection counting. IP anonymisation severely degrades the security function of JA4proxy and should only be used when legally required, with operator acknowledgement of the security trade-off.

PCI-DSS Considerations

JA4proxy contributes to the following PCI-DSS v4.0 requirements when deployed in front of a cardholder data environment:

Requirement	Control	JA4proxy contribution
1.2	Network access controls	Enforces perimeter access, and TLS fingerprint
1.3	Network access restriction	Blocks traffic from known bad actors (Spamhaus, blacklisted)
6.4	WAF/bot protection	TLS fingerprint analysis tools and scanners
6.5	Change management	Config audit log (manages changes)
10.2	Audit logs	Structured JSON logs with action, IP, and reason
10.3	Audit log protection	Logs shipped to Loki; provides own logs
10.5	Log retention	Log retention policy (set via config)
12.10	Incident response Security monitoring via anomaly detection and alerting	

Table 74: PCI-DSS v4.0 control contributions

SOC 2 Audit Trail Completeness

For SOC 2 Type II assessments, auditors typically request evidence of:

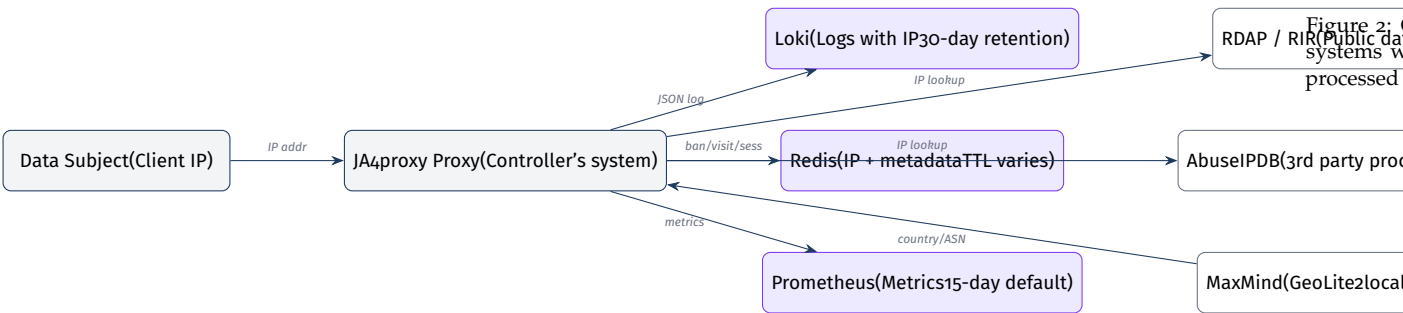
Access control decisions Every connection produces a JSON log entry with action, reason, and client identification. The structured log format provides a complete, queryable audit trail.

Configuration change management The `management:policy_audit` Redis list records every policy change with timestamp, operator attribution, and old/new values. This list is limited to 1000 entries — for long-term retention, ship audit log entries to an external store.

Security monitoring Prometheus metrics, Grafana dashboards, and Alertmanager alert rules provide evidence of continuous security monitoring.

Incident response The analytics node’s campaign detection and score drift alerting provide evidence of automated anomaly detection.

GDPR Data Flow Diagram



Third-Party Data Processors

JA4proxy sends client IP addresses to the following third-party services. Operators must ensure appropriate data processing agreements (DPAs) are in place:

Service	Data sent	Retention by processor	DPA required?
AbuseIPDB	Client IP address	Per AbuseIPDB privacy policy	Yes, if processing EU personal data
RDAP/WHOIS (RIRs)	Client IP address	Public records; varies by RIR	Assess per RIR
MaxMind GeoLite2	None (local DB)	N/A — local file, no network calls	No
DNS resolver	Reverse DNS query for client IP	Per resolver retention policy	Depends on resolver

Disabling AbuseIPDB for Privacy Compliance

If your jurisdiction or DPA requires that client IP addresses not be sent to third parties without consent, set `abuseipdb.enabled: false`. This removes the only signal that requires active transmission of client IPs to an external service. RDAP queries can also be disabled (`rdap.enabled: false`). MaxMind GeoLite2 operates

entirely on a local database file with no network calls.

Glossary

TERMS ARE LISTED ALPHABETICALLY. Cross-references point to the chapter or section where each concept is covered in detail.

Action decider

The final pipeline layer (Layer 14) that maps a computed risk score and the current dial setting to one of the six actions: allow, flag, rate_limit, tarpit, block, or ban. The action decider applies the operator's dial policy to the objective score produced by the risk scorer. See Chapter , Section on the Action Decider.

ALPN (Application-Layer Protocol Negotiation)

A TLS extension (extension type 0x0010) in the ClientHello that lists the application-layer protocols the client supports, in preference order. The most common values for HTTPS are h2 (HTTP/2) and h1 (HTTP/1.1). JA4proxy uses ALPN to identify browser traffic — real browsers always negotiate h2 or h1. See Chapter , Layer 1.

ASN (Autonomous System Number)

A globally unique number assigned to a network operator (ISP, cloud provider, university, etc.) by a Regional Internet Registry (RIR). IP prefixes are announced under an ASN in the global BGP routing table. JA4proxy uses the MaxMind GeoLite2 ASN database to map client IPs to their originating ASN and organisation name. See Chapter , Signal 4.

AbuseIPDB

A community-maintained reputation database of IP addresses reported for malicious activity (scanning, brute-force, spam, C2 traffic). Each IP receives a confidence score 0–100. JA4proxy caches AbuseIPDB responses in Redis and uses the score as a weighted signal. Scores below 50 are suppressed to avoid false positives from shared infrastructure. See Chapter , Signal 7.

Attacker attribution

Pipeline Layer 11. The process of correlating JA4, JA4X, and JA4T fingerprints across multiple source IPs to identify coordinated campaigns, botnet activations, and IP-rotation evasion attempts. Attribu-

tion findings are written by the analytics node and read back by the pipeline. See Chapter , Section on Layer 11.

Asymmetry principle

The core design principle of JA4proxy: the cost of a false positive (real user blocked) is higher than the cost of a false negative (bad bot allowed). This principle governs all threshold defaults, caching TTLs, and fallback behaviours. When in doubt, fail open. See Chapter , Section on Design Principles.

Ban TTL

The time-to-live of an IP ban entry in Redis (`ban:{ip}`). Initial TTL is configurable (default 300 seconds); it escalates by doubling on each re-ban up to a maximum of 604800 seconds (7 days). After the TTL expires, the ban entry is automatically removed by Redis and the IP is re-evaluated on next connection. See Chapter , Section on Blocking Escalation.

Beaconing

A network traffic pattern characterised by regular, periodic connections with low variance in inter-arrival times (IAT). Beaconing is characteristic of C2 (command and control) implants and automated scheduled tasks, as opposed to human browsing which has highly variable timing. Detected using the coefficient of variation of IAT. See Chapter , Signal 6.

Bloom filter

A probabilistic data structure that supports membership queries with a controlled false positive rate and no false negatives. JA4proxy uses Redis Bloom filters (via RedisBloom module) for enrichment deduplication — to avoid re-querying AbuseIPDB or RDAP for IPs that have already been processed. If RedisBloom is unavailable, a Redis SET with a 24h TTL is used as fallback. See Chapter .

Bypass condition

A fast-path rule that short-circuits the pipeline immediately when a specific condition is met, without running the signal collection or scoring layers. JA4proxy has eight bypass conditions, each independently configurable. See Chapter , Section on the Fast-Path Bypass System.

CIDR (Classless Inter-Domain Routing)

IP prefix notation combining a base address with a prefix length (e.g. 185.220.96.0/21). Used in JA4proxy for subnet blocking, trusted upstream CIDR verification, and block expansion. CIDR matching is always performed in-process using a trie data structure — never via Redis queries. See Chapter .

ClientHello

The first message sent by a TLS client in a TLS handshake. It is transmitted in plaintext (TLS record type 0x16, content type 0x01) before any encryption is established. It contains the TLS version, cipher suite list, extensions (including SNI, ALPN, supported groups), and other parameters. JA4proxy reads and parses the ClientHello without participating in the TLS handshake. See Chapter , Section on Data Flow.

Cold cache

The state of the local in-process decision cache immediately after startup, or for a newly seen client (a client IP + JA4 fingerprint not seen recently). Cold cache state means there is no locally cached decision; the pipeline must execute in full (including Redis lookups and signal collection) to produce a decision for this client. Contrast with *warm cache*.

Coefficient of variation (CV)

A statistical measure of dispersion: $CV = \text{standard deviation} / \text{mean}$. Used in the beaconing detector to quantify the regularity of inter-arrival times. Low CV (<0.4) indicates regular, beaconing-like timing. High CV (>0.8) indicates variable, human-like timing. See Chapter , Signal 6.

Dial

A scalar integer setting (0–100) that controls how aggressively risk scores are translated into blocking actions. At dial=0, the proxy is in monitor-only mode — all connections are allowed regardless of score. At dial=100, configured thresholds apply fully. The dial should be raised incrementally after validating the score distribution. See Chapter , Section on dial.

DGA (Domain Generation Algorithm)

An algorithm used by malware to generate pseudo-random domain names for C2 communication, making blacklisting difficult. DGA-generated domains typically have high Shannon entropy and use unusual TLD patterns. JA4proxy detects DGA-like SNI values as a scored signal. See Chapter , Signal 2.

FCrDNS (Forward-confirmed reverse DNS)

A DNS validation technique: the PTR record for an IP resolves to a hostname, and that hostname's A/AAAA record resolves back to the original IP. FCrDNS is a lightweight indicator of network infrastructure legitimacy — residential ISPs, major cloud providers, and CDNs maintain consistent FCrDNS. See Chapter , Signal 5.

Fail open

The principle that in all uncertainty or failure conditions, connections

are allowed rather than blocked. An unavailable external service, a Redis error, or an unrecognised fingerprint all result in a zero signal contribution — the score can only decrease when services fail, not increase. The fail-open guarantee is fundamental to JA4proxy's false positive minimisation. See Chapter , Section on the Fail-Open Guarantee.

Fast path

The set of pipeline layers (0–2) that short-circuit immediately without running the signal collection or scoring layers. Fast-path decisions (from local cache, bypass conditions) complete in under 100 microseconds. Connections that bypass the fast path proceed to the signal collection zone. See Chapter .

HyperLogLog

A probabilistic data structure for counting distinct elements with fixed memory cost (12KB per key in Redis). Used in JA4proxy to count unique IPs per /24 (IPv4) or /48 (IPv6) CIDR prefix. Standard error is 0.81%. Individual elements cannot be removed from a HyperLogLog without clearing the entire structure. See Chapter .

IAT (Inter-arrival time)

The time elapsed between consecutive connections from the same source (IP + JA4 fingerprint pair). IAT analysis is the core of the beaconing detection algorithm. IAT values are stored as timestamps in a Redis Sorted Set with a 30-minute sliding window. See Chapter , Signal 6.

JA4

A TLS client fingerprinting algorithm that produces a deterministic hash from the TLS ClientHello: cipher suites (sorted), TLS extensions (sorted), ALPN values, and signature algorithms. Produces a string of the form `t13d1516h2_8daaf6152771_02713d6af862`. The algorithm was developed by FoxIO and is defined at <https://github.com/FoxIO-LLC/ja4>. See Chapter .

JA4X

The extended JA4 fingerprint that includes the TLS extension type and ordering information — not just which extensions are present, but in what order the client sent them. JA4X can distinguish clients that share the same JA4 value but have different extension ordering (e.g., a real Chrome vs a JA4-copying tool). See Chapter .

JA4T

The TCP transport fingerprint component of the JA4 family. Encodes the TCP window size, TCP options (MSS, window scale, SACK, timestamps), and their ordering. JA4T fingerprints the OS-level TCP/IP stack, making it harder to spoof than JA4 (which fingerprints the TLS

library layer). See Chapter , Signal 3.

Monitor mode

The operating mode when `dial: 0`. In monitor mode, the full pipeline runs (all signals collected, scores computed, actions decided), but every connection is allowed regardless of score. Actions are logged as `MONITOR`. Monitor mode is the default and recommended starting point for all deployments. See Chapter , Section on the Action Decider.

mTLS (Mutual TLS)

TLS authentication where both the server and the client present certificates. A valid mTLS client certificate is cryptographic proof of identity — the client holds the private key corresponding to a certificate signed by the configured trusted CA (`config/trusted_cas.pem`). By default, mTLS clients bypass the scoring pipeline entirely. See Chapter , Layer 1c.

PROXY protocol

A binary or text framing format prepended to a TCP stream by a load balancer to convey the original client IP address and port. JA4proxy requires HAProxy to use PROXY protocol v2 (binary format) so that the real client IP is available after the TCP connection arrives from HAProxy. See Chapter , Section on Stage 1.

RDAP (Registration Data Access Protocol)

The modern replacement for WHOIS. Provides structured JSON responses about IP address registrations from Regional Internet Registries (RIRs: ARIN, RIPE, APNIC, LACNIC, AFRINIC). JA4proxy uses RDAP to identify the organisation owning an IP block and cross-reference it against the known-bad organisations list. See Chapter , Signal 8.

Risk score

A scalar integer in the range 0–100 produced by the risk scorer (pipeline Layer 13) by confidence-weighted aggregation of all signal contributions. A score of 0 means clean; 100 means maximum risk. The `dial` setting controls how the score is translated into a blocking action. See Chapter , Section on the Risk Scorer.

Signal

A scored observation produced by one of the eight signal checks (pipeline Layers 3–10). Each signal returns a `RiskSignal(score, weight, reason)` object. Signals run concurrently via `asyncio.gather()`. A signal that cannot produce a result (service unavailable) returns `score=0` (fail open). See Chapter .

SNI (Server Name Indication)

A TLS extension (type 0x0000) in the ClientHello that specifies the hostname the client intends to connect to. SNI allows a single server to host multiple TLS virtual hosts on the same IP. JA4proxy uses SNI to detect missing SNI, IP-literal SNI, and DGA-like domain names. See Chapter , Signal 2.

Spamhaus DROP/EDROP

The Spamhaus Don't Route Or Peer (DROP) and Extended DROP (EDROP) blocklists contain IP ranges that have been hijacked or directly allocated to cybercriminals. False positive rate is near zero. JA4proxy downloads these lists at startup and on a refresh interval, storing them in an in-process trie for $O(\log n)$ lookup. See Chapter , Layer oe.

Tarpit

An action that accepts a connection but delivers data to the backend at an artificially throttled rate, consuming the attacker's connection slots and CPU while expending minimal resources on the defender's side. Effective against connection-limited scanners and brute-force tools. JA4proxy implements tarpit as a rate-controlled TCP relay. See Chapter , Section on the Six Actions.

TLS fingerprinting

The technique of computing a deterministic identifier from the parameters in a TLS ClientHello (cipher suites, extensions, ALPN, etc.) to identify the TLS client implementation. Different applications produce different fingerprints even when connecting to the same server. The JA4 algorithm is the fingerprinting standard used by JA4proxy. See Chapter .

Warm cache

The state of the local in-process decision cache when it contains a recent decision for a given client — keyed on the client IP together with its JA4 fingerprint, so clients sharing a fingerprint do not share a decision. A warm cache hit means the pipeline can return a decision in under 100 microseconds without any Redis or signal collection overhead. Cache entries have asymmetric TTLs: ALLOW entries are cached for 30 minutes; every blocking action for 30 seconds (ADR-003). Contrast with *cold cache*.

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What JA4proxy does

JA4proxy intercepts the TLS ClientHello — which is always transmitted in plaintext, before encryption begins. It computes a JA4 fingerprint from the cipher suites, extensions, ALPN, and elliptic curves advertised by each client, then routes the connection through a multi-signal scoring pipeline to decide: *allow, tarpit, block, or ban*. Allowed connections are forwarded byte-for-byte unchanged. The proxy never decrypts, never holds your private keys, and is transparent to both client and backend.

Key facts

- 0% false positive rate on browser traffic — by design
- 94–99% of malicious traffic blocked in testing
- Default dial = 0: monitor mode, never blocks on first deploy
- All bans carry a 5-minute TTL — false positives self-heal
- Never decrypts: no key custody, no SSL inspection compliance risk
- Scales horizontally; Redis synchronises bans across all instances

Resources

Source code & issue tracker: github.com/seanpor/JA4proxy

JA4 fingerprint specification: github.com/FoxIO-LLC/ja4

Documentation index: [docs/README.md](#) in the repository